Indirect reciprocity with inertial reputation updating

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ABSTRACT

Indirect reciprocity explains cooperation among strangers, because individuals who help others gain a good reputation and are helped in return. This mechanism works well when reputations are public. When observers form their opinions privately, however, errors create disagreements that can undermine cooperation. Most models also let each new assessment replace the previous reputation, so that a single error can overturn an established standing. Here we introduce inertial reputation updating, in which evidence for losing and for regaining standing accumulates along separate paths, and incomplete evidence can already affect treatment. Using simulations and analytical calculations, we show that the allocation of inertia between these paths, rather than its total amount, determines whether private reputations are both accurate and shared. Compact, norm-specific architectures, selected solely on these reputational criteria, sustain stable and highly cooperative outcomes under four classical social norms and enlarge their basins of attraction. Tolerance of errors, however, can also shelter occasional defectors. Letting early warning states carry partial consequences removes this vulnerability for the opportunistic strategies we consider, while keeping cooperation high. Reputation-based cooperation thus depends not only on which behaviors are judged good, but also on how reputations accumulate and express behavioral evidence.

Introduction

Cooperation among unrelated individuals is sustained in part by reputation[1,2]. By helping others, individuals improve their standing and become more likely to receive help from third parties, a mechanism known as indirect reciprocity[3–6]. Theory has identified social norms that make reputation-based cooperation evolutionarily stable[7,8], most notably the leading-eight norms, which sustain cooperation when reputations are public, even in the presence of occasional errors[8,9]. These results rest on agreement about who is good and who is bad. When reputations are private and assessments are noisy, individuals hold different views of the same person[10,11], and most leading-eight norms no longer maintain stable cooperation[12–15]. A central problem for indirect reciprocity is therefore how reputations can remain both accurate and shared under private assessment. They must distinguish those who cooperate from those who do not, and different observers must largely agree about who is who[16,17].

Part of this fragility lies in how reputations are updated. In the standard binary framework, each new assessment replaces the previous image of an individual[8,12]. An isolated error can therefore reverse a reputation built over many interactions, and the next observation can reverse it again. Under private assessment, such reversals do not remain local. They amplify disagreements among observers and feed back into behavior. A discriminator who refuses to help a recipient it regards as bad may itself be condemned by an observer who regards that recipient as good[12,18,19]. Recent work has made reputation systems more robust to errors and disagreement in two ways. Some mechanisms restore agreement among observers, through public institutions[20], gossip[21], empathetic evaluation[18] or the synchronization of private opinions[17,22,23]. Others use more information before a reputation changes, through generous or stochastic assessment[24,25], graded or quantitative reputations[26–29], reasoning about errors[30,31] and the aggregation of several observations[32]. Most of these mechanisms share a tacit assumption. Losing and regaining standing are governed by the same rule, such as a single score, threshold or observation window[28,32,33].

Many human institutions do not make this assumption. Communities that successfully manage shared resources sanction first offences mildly and escalate only after repeated violations[34,35], and a criminal record stops signaling elevated risk only after years without a further offence[36]. These practices treat losing and regaining standing as separate processes and let incomplete evidence carry partial consequences. Relaxing the same assumption in models of indirect reciprocity raises two questions. First, should losing a good reputation require as much evidence as regaining one? If observers forgive too little, honest mistakes trigger cascades of disagreement. If they restore standing too readily, defectors escape the consequences of their behavior. Which of these risks dominates may depend on the norm that defines good conduct. Second, how should evidence that is accumulating, but not yet conclusive, affect how someone is treated? Waiting for conclusive evidence protects cooperators from errors, but the same forbearance may shelter individuals who defect deliberately, provided they do so only occasionally[24,32]. Whether a reputation system can forgive honest mistakes without excusing deliberate defection therefore remains open.

Here, we develop a model of indirect reciprocity in which reputations have inertia (Fig. 1). An established reputation changes only after contrary evidence has accumulated, and the amount of evidence required can differ between losing and regaining standing. Negative evidence moves an individual in good standing through a sequence of warning states before the reputation turns bad, and positive evidence moves an individual in bad standing through recovery states before it turns good. The length of each path sets the inertia in that direction, and each intermediate state can already affect how the individual is treated. The framework thus separates two design choices of a reputation system, namely how much evidence is needed to change an established reputation in each direction and how strongly incomplete evidence is expressed in the meantime.

We find that the allocation of inertia between the two directions, rather than its total amount, determines whether private observers agree with one another and still distinguish cooperators from defectors. The beneficial direction depends on the norm. Compact architectures, selected solely by these reputational criteria, make cooperation stable and easy to reach under each of four classical social norms (Scoring, Shunning, Stern Judging and Simple Standing). At the same depth, inertia placed on the wrong path can leave no cooperation at all. We further show that the tolerance that protects cooperators can also shelter opportunists who defect only occasionally, even when unconditional defectors are excluded. Allowing warning states to carry partial consequences closes this loophole for the opportunistic strategies we consider, while cooperation remains high. Reputation-based cooperation therefore depends not only on which behaviors a society judges as good, but also on how readily it revises those judgments in each direction and on how it treats individuals before a judgment changes.

Results

A framework for inertial reputation updating

We consider a well-mixed population in which randomly matched donors and recipients play the donation game. A donor can help a recipient at a cost c, conferring a benefit $b > c$. Unconditional cooperators (ALLC) always intend to help, unconditional defectors (ALLD) never do, and discriminators (DISC) help only recipients that they regard as good. Reputations are private[12,13]. Each observer keeps its own record of every individual and judges a donor's action by applying a social norm to its own view of the recipient. We study the four second-order norms Scoring (SC), Shunning (SH), Stern Judging (SJ) and Simple Standing (SS)[8]. All four judge helping a good recipient as good and refusing such help as bad. They differ in how they judge actions towards recipients regarded as bad. Intended cooperation fails with probability e_1, and observers record the opposite assessment with probability e_2 (Methods).

Under instantaneous updating, each new assessment replaces the observer's previous image of the donor (Fig. 1a). Inertial updating instead lets evidence accumulate before an established reputation changes (Fig. 1b). Consecutive negative assessments move an individual in good standing through k_1 warning states, $\mathrm{W}_1,\ldots,\mathrm{W}_{k_1}$, before the record turns bad. Consecutive positive assessments move an individual in bad standing through k_2 recovery states, $\mathrm{S}_1,\ldots,\mathrm{S}_{k_2}$, before the record turns good. An assessment that supports the established judgment returns the record to the corresponding established state. Losing good standing therefore requires k_1+1 consecutive negative assessments, and regaining it requires k_2+1 consecutive positive ones. We call k_1 the tolerance depth, k_2 the strictness depth and $K = k_1 + k_2$ the total depth. Because the two depths are set independently, losing and regaining standing need not be equally difficult. The pair $(k_1,k_2) = (0,0)$ recovers instantaneous updating.

An internal record must still be translated into treatment. Whenever a record is consulted, either to decide on an action or to assess another individual's action, its state is interpreted as a good or bad image. Warning state W_m is interpreted as good with probability α^m, and recovery state S_n as bad with probability β^n. We call α the tolerance strength and β the strictness strength. With $\alpha = \beta = 1$, an established image persists until the corresponding path is completed. With smaller values, incomplete evidence already affects treatment. The depths thus determine how much evidence is needed to change a reputation, whereas the strengths determine how strongly incomplete evidence is expressed in the meantime. Throughout, we compare instantaneous updating, full one-step inertia ($k_1 = k_2 = 1$ with $\alpha = \beta = 1$) and norm-specific multi-step architectures with $\alpha = \beta = 1$, which we identify next.

Directional inertia shapes private reputation dynamics

A useful reputation system must achieve two things at once. Observers should agree about individual targets, and their judgments should distinguish cooperative from defective strategies. We examined both properties in finite populations with equal numbers of ALLC, ALLD and DISC individuals, in which each DISC observer keeps a separate record of every individual and strategies do not change (Methods). We measure recognition by the reputational separation $\Delta g = (g_{\mathrm{DISC}} + g_{\mathrm{ALLC}})/2 - g_{\mathrm{ALLD}}$, where g_I is the probability that a DISC observer regards a type-I individual as good. We measure agreement by the mean disagreement $\bar{d}$, the probability that two DISC observers hold different images of the same individual.

Without inertia, every norm fails in a characteristic way (Fig. 2a–d, top row). Under SC, observers are divided about discriminators. SH drives almost all reputations towards bad, SJ sustains widespread disagreement and SS extends trust to

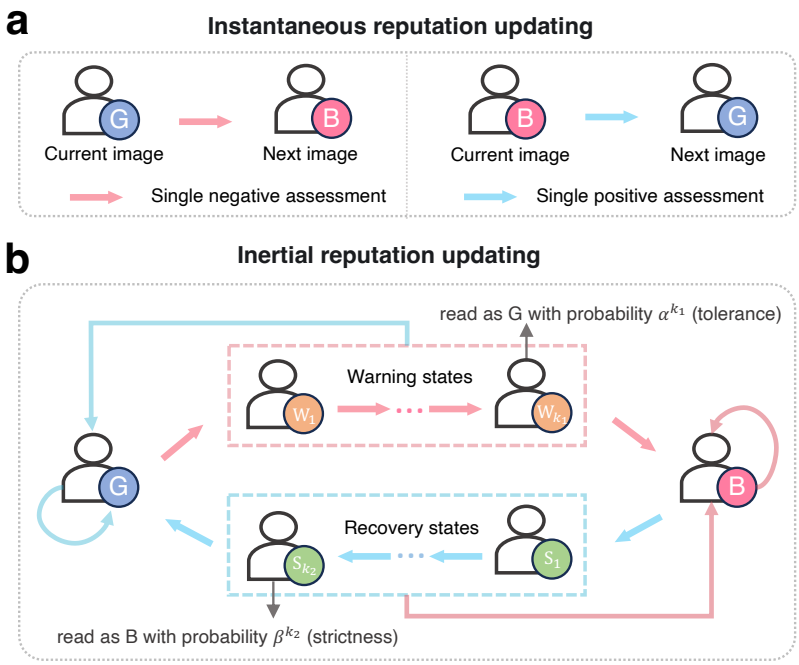


Figure 1. A framework for inertial reputation updating. a, Under instantaneous updating, a single negative or positive assessment replaces the current binary image. **b**, Under inertial updating, negative assessments move a good record through warning states $W_1, \ldots, W_{k_1}$ towards bad, whereas positive assessments move a bad record through recovery states $S_1, \ldots, S_{k_2}$ towards good. An assessment in the opposite direction returns the record to the corresponding established state. The tolerance depth k_1 and the strictness depth k_2 are set independently. Warning state W_m is interpreted as good with probability α^m (tolerance strength α), and recovery state S_n is interpreted as bad with probability β^n (strictness strength β). The diagram labels these probabilities for the final state of each path. Pink and blue arrows denote negative and positive assessments, respectively.

defectors. Full one-step inertia does not resolve these failures (Fig. 2a–d, middle row). Under SH, for example, observers now agree, but only because they regard nearly everyone as bad. Agreement alone is therefore not sufficient.

Both requirements can be met when inertia is allocated in a norm-specific direction (Fig. 2e–h). Allocations with the same total depth produce markedly different separation and disagreement. For each norm, we selected the qualifying allocation with the smallest total depth, using $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$ as operational criteria. This yields $(k_1, k_2) = (2, 0)$ for SC, $(3, 0)$ for SH, $(4, 3)$ for SJ and $(1, 2)$ for SS. Inertia therefore protects good standing under SC and SH, acts in both directions under SJ and slows recovery under SS. With these architectures, observers regard DISC and ALLC as good and ALLD as bad, and they largely agree with one another (Fig. 2a–d, bottom row). Replicated simulations confirm the selection, and the directional pattern persists across a range of criteria (Extended Data Fig. 1 and Supplementary Table S10). These selected multi-step architectures are chosen from reputation statistics alone, without reference to evolutionary outcomes.

Inertia also changes reputation dynamics within a population of discriminators (Extended Data Fig. 2a–d). With one-step inertia, homogeneous DISC populations under SC, SH and SJ are bistable, so their reputation depends on their history. The selected architectures remove this bistability and leave a single high-reputation state. The two depths and the two strengths jointly determine these reputation states (Extended Data Fig. 3). Directional inertia thus allows private observers to form reputations that are both discriminating and consistent. We next ask whether such reputations allow cooperation to evolve.

Directional inertia reshapes strategy dynamics

We let ALLC, ALLD and DISC evolve under the replicator dynamics[37], assuming that reputations equilibrate faster than strategy frequencies change (Methods). At each strategy composition, payoffs are computed from the stationary private reputations reached from initially good records. We report intended cooperation at individual equilibria and the mean realized cooperation $\bar{\rho}$, averaged over initial strategy compositions. Without inertia, SH and SJ do not support cooperation, and under SC and SS a pure DISC population is not a stable cooperative state (Fig. 3a). Full one-step inertia makes highly cooperative DISC populations locally stable under all four norms (Fig. 3b). Under SH and SJ, however, their basins of attraction remain narrow, so cooperation is reached from only a small fraction of initial compositions.

The selected multi-step architectures make cooperation both stable and accessible (Fig. 3c). Under SC, SH and SJ, a pure DISC population is a stable equilibrium with near-complete intended cooperation. It is reached from 70–88% of initial

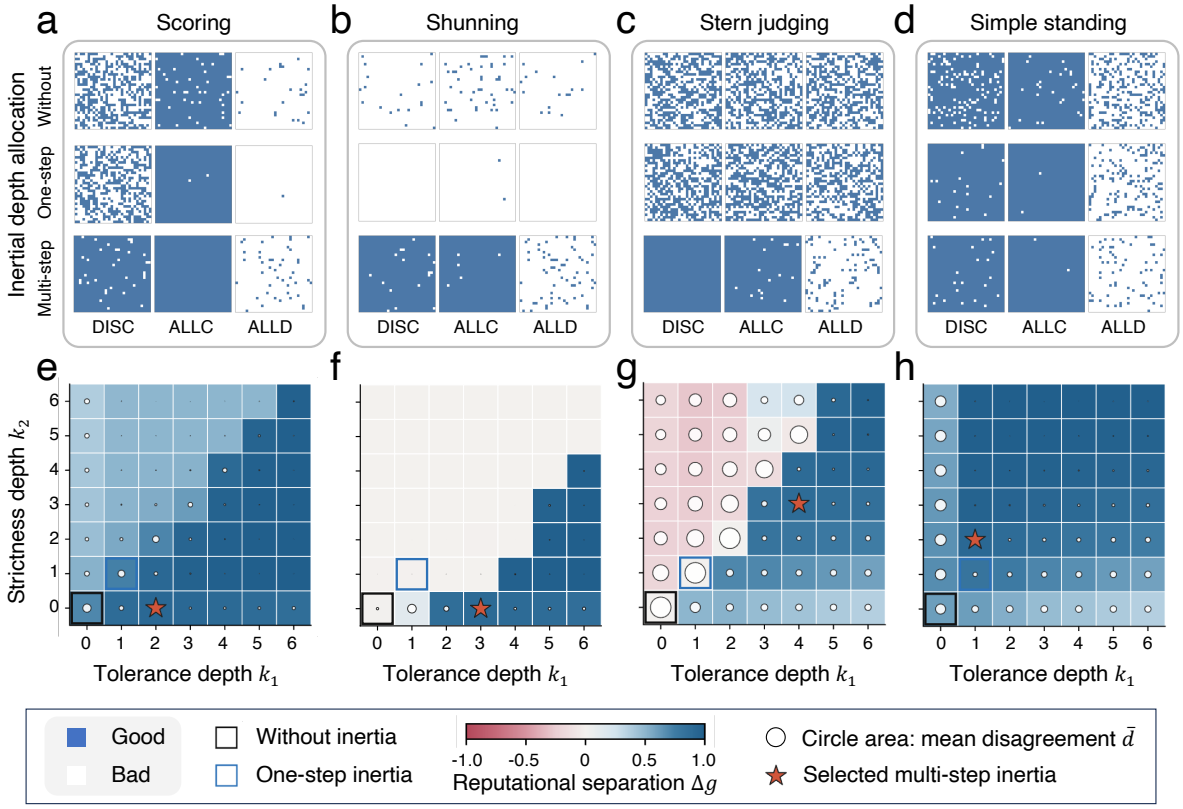


Figure 2. The allocation of inertia shapes private agreement and strategy recognition. a–d, Illustrative private-reputation matrices under Scoring (SC; **a**), Shunning (SH; **b**), Stern Judging (SJ; **c**) and Simple Standing (SS; **d**). Panel rows show instantaneous updating (without inertia), full one-step inertia and the selected multi-step architectures. Within each matrix, rows are DISC observers and columns are DISC, ALLC and ALLD targets. Blue and white indicate good and bad images. Each matrix shows the final state of a single run. **e–h**, Separation Δg (color) and mean observer disagreement $\bar{d}$ (circle area) across tolerance depth k_1 and strictness depth k_2. Black and blue outlines mark instantaneous updating $(0,0)$ and one-step inertia $(1,1)$, respectively. Stars mark the pairs with the smallest total depth K that satisfy $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$. Parameters: $e_1 = e_2 = 0.02$, equal proportions of the three strategies, population size $N = 90$ for the matrices and $N = 150$ for the depth scans (one run per condition), and $\alpha = \beta = 1$ in the depth scans.

compositions, compared with 8% and 23% under one-step SH and SJ. Under SS, the cooperative attractor is a stable mixture of DISC and ALLC. Most of the resulting gain in mean cooperation comes from the enlarged basins rather than from higher cooperation within them (Supplementary Note 1.5). The ALLD–DISC edge of the simplex shows why the basins expand (Extended Data Fig. 4). Under one-step inertia, discriminators lose their good reputation abruptly once they become sufficiently rare, and with it their payoff advantage over ALLD. The selected architectures shift this threshold to lower DISC frequencies or replace it with a smooth transition, so discriminators keep their advantage when they are rarer.

Whether an established DISC population resists rare unconditional mutants has a compact answer. Let r_{R}^* denote the probability that a resident is regarded as good. When both denominators are positive, pure DISC resists rare ALLD and ALLC mutants if and only if

$$R_{\mathrm{L}} \equiv \frac{r_{\mathrm{R}}^*}{r_{\mathrm{R}}^* - \mathscr{R}\left[r_{\mathrm{R}}^* P_{\mathrm{D|G}}^+ + (1-r_{\mathrm{R}}^*) P_{\mathrm{D|B}}^+\right]} < \frac{b}{c} < \frac{1-r_{\mathrm{R}}^*}{\mathscr{R}\left[r_{\mathrm{R}}^* P_{\mathrm{C|G}}^+ + (1-r_{\mathrm{R}}^*) P_{\mathrm{C|B}}^+\right] - r_{\mathrm{R}}^*} \equiv R_{\mathrm{U}}. \tag{1}$$

Here $P_{\mathrm{A|Y}}^+$ is the probability that an observer assesses intended action $A \in \{\mathrm{C},\mathrm{D}\}$ positively when it regards the recipient as $Y \in \{\mathrm{G},\mathrm{B}\}$, including both kinds of error (Methods). Because mutants are rare, their recipients are residents, and each bracketed argument is the overall probability u that the mutant's action is assessed positively. The reputation map $\mathscr{R}(u)$ gives the stationary probability that an inertial record is interpreted as good when each assessment is positive with probability u (Methods). The lower bound requires ALLD to lose enough help to offset the cost it saves. The upper bound prevents ALLC from gaining more through a better reputation than it pays for also helping recipients regarded as bad. Without inertia, $\mathscr{R}(u) = u$ and no benefit-to-cost ratio satisfies both bounds (Supplementary Note 2.2). Inertia opens this window, and the selected architectures widen it considerably under SC, SH and SJ. Under SS, the upper bound of the selected architecture lies at

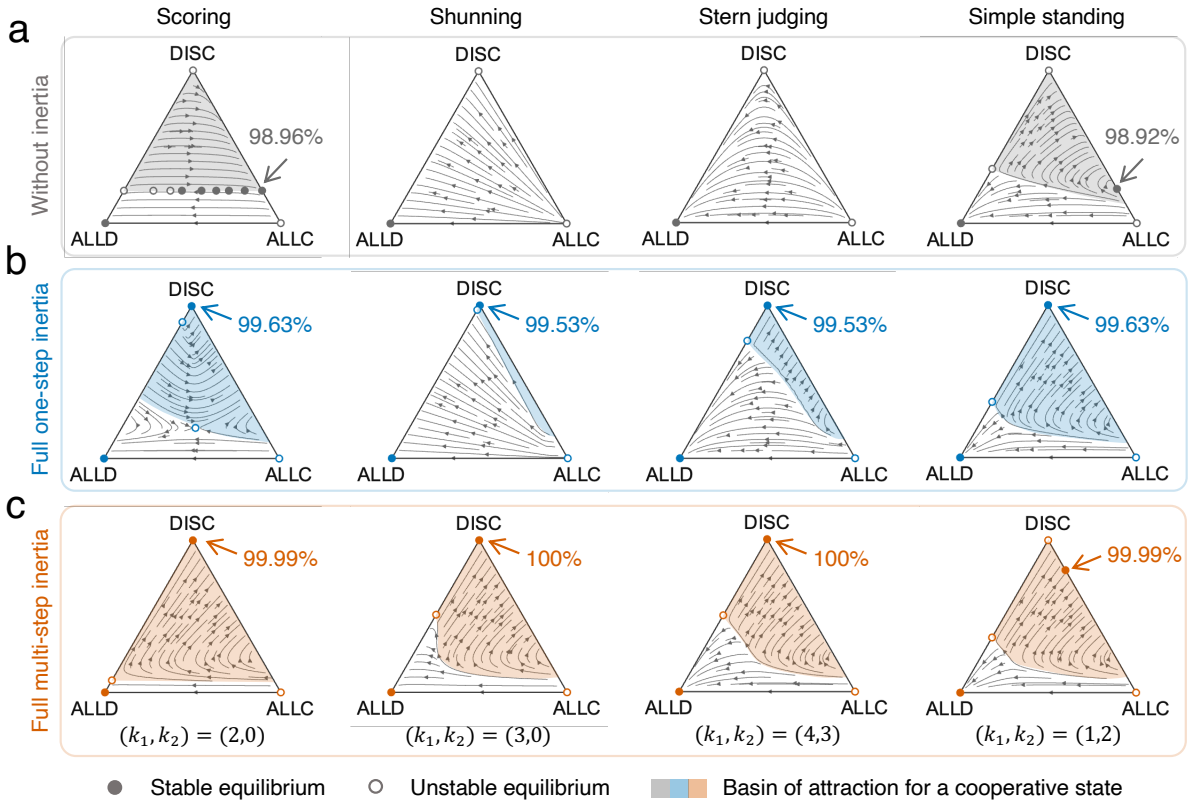


Figure 3. Directional inertia reshapes the evolutionary dynamics of cooperation. a–c, Evolutionary dynamics of ALLC, ALLD and DISC under SC, SH, SJ and SS (columns) without inertia (**a**), with full one-step inertia (**b**) and with the selected multi-step architectures (**c**). Arrows indicate the direction of evolution. Filled and open circles denote stable and unstable equilibria, respectively. For equilibria on an edge of the simplex, markers refer to stability along that edge (Methods). Percentages give intended cooperation at the indicated equilibria, and shading marks the basin of attraction of a cooperative state. Without inertia, SC has a line of equilibria on which all three payoffs are equal (Supplementary Note 1.5). The selected depths (k_1, k_2) are $(2,0)$, $(3,0)$, $(4,3)$ and $(1,2)$ for SC, SH, SJ and SS, respectively. Reputations are relaxed from initially good records at every composition. Parameters: $b/c = 5$, $e_1 = e_2 = 0.02$ and $\alpha = \beta = 1$. [Draft: the SC panel in a will be redrawn, with the line of equilibria shown as a single line (solid where it attracts, dashed where it repels).]

$b/c \approx 4$, below the reference ratio $b/c = 5$. ALLC can therefore invade pure DISC, and the population settles in the highly cooperative DISC–ALLC mixture instead of losing cooperation.

These conclusions hold beyond the reference conditions. The selected architectures reach similar cooperation from all-good and all-bad initial reputations, whereas one-step inertia under SH and SJ loses cooperation when records start bad (Extended Data Fig. 2e–h). They support more cooperation than instantaneous updating and one-step inertia across most benefit-to-cost ratios (Extended Data Fig. 5). At one-step depths, tolerance and strictness strengths have distinct, norm-dependent effects on strategy dynamics (Extended Data Fig. 6). Across implementation and assessment errors, the selected architectures exclude ALLD over a much wider range than one-step inertia under SH and SJ (Extended Data Fig. 7). Directional inertia thus turns reliable private reputations into an evolutionary advantage for cooperation. Because the selected architectures differ in both total depth and allocation, we next separate these two factors.

The direction of inertia matters at a fixed depth

We compared all allocations (k_1, k_2) with the same total depth K (Fig. 4). At a given depth, cooperation depends strongly on allocation. Under SC with eight intermediate states, for example, $\bar{p}$ ranges from zero when all states are recovery states to high cooperation when all are warning states. The most cooperative allocations protect good standing under SC and SH, combine both paths under SJ at larger depths, and pair a short warning path with a longer recovery path under SS. The full plane of depths shows the same patterns (Extended Data Fig. 8). Notably, the four architectures selected from reputation statistics alone are the most cooperative allocations at their respective depths. The two reputational criteria therefore identify compact architectures that also perform well in evolution.

More depth does not necessarily yield more cooperation. Under SC, the most cooperative allocation gains little beyond

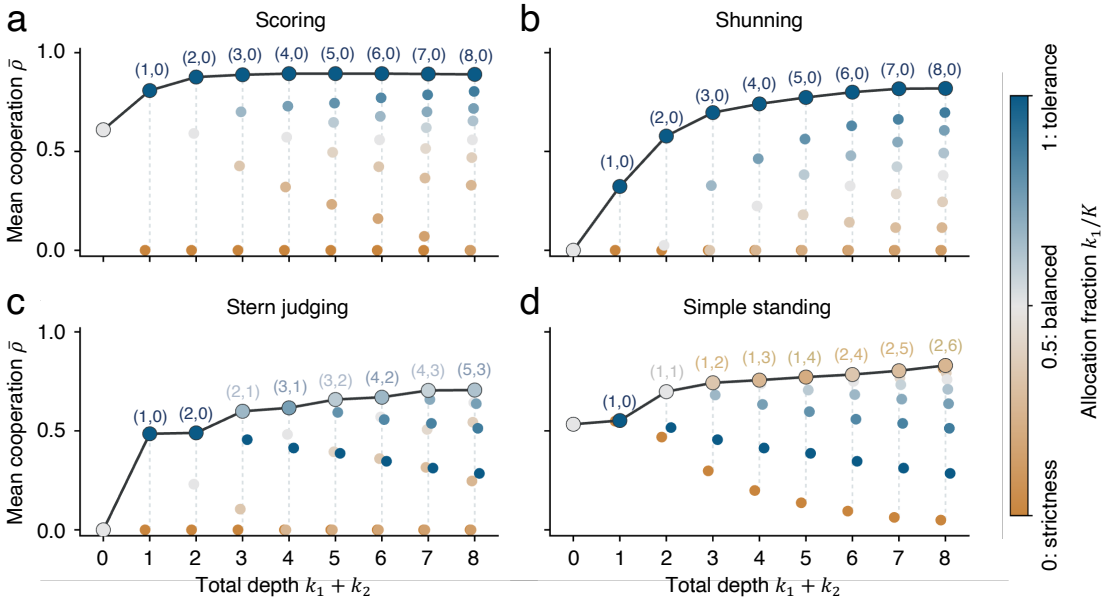


Figure 4. More inertia does not always yield more cooperation. a–d, Mean realized cooperation $\bar{\rho}$ under SC (**a**), SH (**b**), SJ (**c**) and SS (**d**) as a function of total depth $K = k_1 + k_2$. Each point represents one allocation (k_1, k_2), with color indicating the tolerance fraction k_1/K. $K = 0$ is the instantaneous baseline. Large outlined points connected by black lines identify the largest sampled mean at each K, with depth pairs labeled. Cooperation is averaged over 512 initial strategy compositions sampled uniformly from the simplex (Methods). Parameters: $b/c = 5$, $e_1 = e_2 = 0.02$ and $\alpha = \beta = 1$. [Draft: this figure was drawn from the pre-verification data. In the verified data the SS envelope labels are (0,1) at $K = 1$ and (1,5) at $K = 6$. Redraw from verified/figure4_ed6/data/figure4.]

three warning states, and further states do not increase cooperation. The value of additional depth depends on the norm and on how the depth is allocated, and a moderate, appropriately allocated depth already supports high cooperation. Protecting good standing, however, means tolerating some negative evidence. We therefore ask whether occasional defectors can exploit this tolerance.

Tolerance creates opportunistic risk, and tuning its strength moderates it

Tolerance protects discriminators from erroneous negative assessments. The same protection might also shelter individuals who defect deliberately but only occasionally. We therefore introduce a rare opportunistic discriminator (O-DISC), which behaves like DISC except that it withholds help from a recipient it regards as good with probability δ, independently in each interaction. The endpoints $\delta = 0$ and $\delta = 1$ correspond to DISC and ALLD. Let $r_X^*(\delta)$ denote the stationary probability that the mutant is regarded as good. Its payoff advantage over residents, in units of the expected cost of an intended donation, is

$$\Lambda(\delta) = \frac{\Pi_X - \Pi_R}{c(1-e_1)} = \delta r_R^* - \frac{b}{c}\left[r_R^* - r_X^*(\delta)\right]. \tag{2}$$

The first term is the cost that the mutant saves, and the second is the benefit that it loses through a worse reputation. A positive Λ permits invasion. Resistance to ALLD establishes only $\Lambda(1) < 0$ and leaves partial opportunism, $0 < \delta < 1$, untested.

Excluding ALLD indeed does not exclude partial opportunism (Fig. 5a). All five SJ mechanisms compared in Fig. 5a exclude ALLD. Nevertheless, under the selected architecture with full tolerance strength, every O-DISC mutant with $0 < \delta < 0.365$ can invade, and one-step inertia admits a narrow window of the same kind. Two alternative mechanisms that also buffer isolated negative evidence, aggregation of two sampled assessments[32] and a bounded quantitative score[28], show similar windows. These mechanisms were neither optimized nor matched in memory to the inertial record. They therefore illustrate a shared vulnerability rather than a ranking (Supplementary Note 2.4).

This vulnerability originates in the reputational response to weak opportunism. At $\delta = 0$, the slope of the invasion payoff is

$$\Lambda'(0) = r_R^*\left[1 - \frac{b}{c}\Delta\Phi^* \mathscr{R}'(u_R^*)\right]. \tag{3}$$

Here $\Delta\Phi^*$ is the difference between the probabilities that intended cooperation and intended defection are assessed positively in the resident population, and $\mathscr{R}'(u_R^*)$ is the slope of the reputation map at the residents' positive-assessment probability u_R^*. Each unit of withheld cooperation lowers the mutant's positive-assessment probability by $r_R^*\Delta\Phi^*$, and the reputation map converts

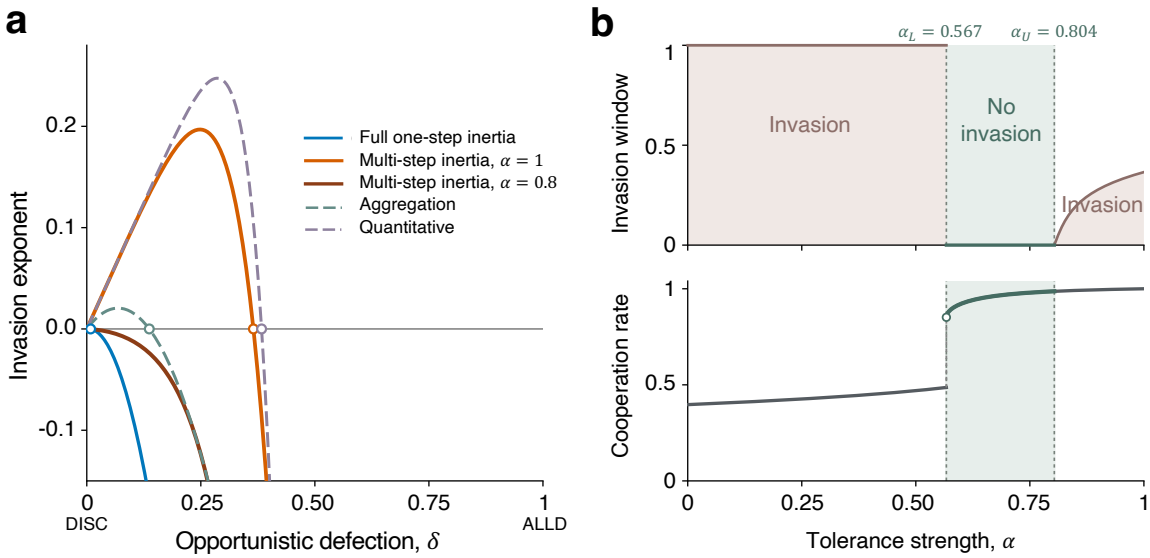


Figure 5. Tolerance can permit opportunistic invasion, and reducing tolerance strength can limit it. a, Normalized payoff advantage Λ of an O-DISC mutant against a DISC resident under SJ, as a function of the omission probability δ ($\delta = 0$, DISC; $\delta = 1$, ALLD). Curves compare full one-step inertia, the selected multi-step architecture $(4,3)$ with $\alpha = 1$ and $\alpha = 0.8$, an aggregation rule[32] and a quantitative score[28]. Positive values indicate invasion, and open circles mark nonzero crossings. **b**, Selected multi-step architecture $(4,3)$ as the tolerance strength α varies. The upper panel shows the largest detected endpoint δ_c of the invasion window, and the lower panel shows resident intended cooperation r_{R}^*. The green band marks the secure range $\alpha_{\mathrm{L}} < \alpha \leq \alpha_{\mathrm{U}}$, with $\alpha_{\mathrm{L}} = 0.567$ and $\alpha_{\mathrm{U}} = 0.804$. Parameters: $b/c = 5$, $e_1 = e_2 = 0.02$ and $\beta = 1$.

this into a loss of reputation (Methods). If the map is flat near the resident state, weak opportunism costs almost no reputation and the saved cost dominates. For any architecture with a warning path, $\mathscr{R}'(u) \to 1 - \alpha$ as $u \to 1$ (Supplementary Note 1.3). Full tolerance strength therefore makes the response nearly flat near a highly reputed resident. Lowering α gives warning states an immediate chance of being interpreted as bad and so restores a marginal penalty.

A negative slope at $\delta = 0$ does not yet exclude larger omission probabilities. We therefore consider the reputation lost per unit of withheld cooperation, $H(\delta) = [r_{\mathrm{R}}^* - r_{\mathrm{X}}^*(\delta)]/(\delta r_{\mathrm{R}}^*)$. Because $\Lambda(\delta) \leq 0$ exactly when $H(\delta) \geq c/b$, the resident excludes the entire O-DISC family if and only if

$$H_{\min} \equiv \min_{0 \leq \delta \leq 1} \frac{r_{\mathrm{R}}^* - r_{\mathrm{X}}^*(\delta)}{\delta\, r_{\mathrm{R}}^*} \geq \frac{c}{b}, \tag{4}$$

where the ratio at $\delta = 0$ is defined by its limit, $\Delta\Phi^* \mathscr{R}'(u_{\mathrm{R}}^*)$. Even the mutant with the smallest reputational penalty must gain nothing.

Lowering the tolerance strength can satisfy this condition while preserving cooperation. For the selected SJ architecture, reducing α from 1 to 0.8 excludes every O-DISC mutant, and residents remain highly cooperative (Fig. 5a). Lowering α too far, however, destroys the high-reputation resident state. Protection therefore holds within a bounded range, $\alpha_{\mathrm{L}} < \alpha \leq \alpha_{\mathrm{U}}$, which here extends from about 0.57 to 0.80 (Fig. 5b). The lower boundary is a fold of the high-reputation state and does not depend on b/c. The upper boundary marks the point at which the weakest opportunists become neutral, and it rises with b/c (Methods).

Protective tolerance strengths exist under all four norms, although their location depends on the norm and the architecture (Extended Data Fig. 9). At $\alpha = 0.8$, all four selected architectures resist the O-DISC family and maintain high resident cooperation, whereas one-step inertia under SH and SJ has already lost its high-reputation state. Because the fold is fixed while the upper boundary rises with b/c, protective ranges shrink at low benefit-to-cost ratios and can vanish, as under SH at $b/c = 2$. These results concern independent, occasional omissions. They do not cover persistent or temporally correlated opportunism, nor invasion by ALLC, which still succeeds against the selected SS architecture.

Accumulating evidence before a reputation changes and responding to that evidence along the way therefore serve different functions. Depth and direction determine whether reputations are accurate and cooperation is accessible, whereas the strength of warning states determines whether tolerance can be exploited.

Discussion

[Expanded Discussion (revision 2), to be revised further.]

Research on indirect reciprocity has largely asked which behaviors a society should regard as good[8,9,38,39]. Our results highlight a second design dimension, namely how a society turns a sequence of observations into a reputation. With the norm held fixed, the way reputations accumulate and express evidence decides whether private observers agree, whether they distinguish cooperators from defectors and whether cooperation can evolve. Separating the loss of standing from its recovery, and the accumulation of evidence from its interpretation, makes this dimension explicit and analytically tractable.

The most consequential property of an inertial architecture is its direction rather than its size. In the four norms we studied, the beneficial direction mirrors the type of error that private assessment produces. Under SC and SH, discriminators who justifiably refuse to help bad recipients are themselves condemned, so cooperative individuals accumulate unwarranted negative assessments. Protecting good standing filters exactly these errors. Under SS, defection against bad recipients is approved, which allows defectors to regain good standing too easily, and slower recovery counteracts this. Under SJ, every verdict depends on the observer's view of the recipient, which exposes cooperators and defectors alike to errors and calls for inertia in both directions. This correspondence suggests that norms and rules for integrating evidence should be analyzed together. Shunning and Stern Judging fail under instantaneous private updating, yet both sustain cooperation once they are paired with an appropriate architecture. Comparisons and evolutionary analyses of norms under private assessment[39,40] may therefore reach different conclusions when reputations have inertia.

Our results also clarify what private reputation systems need to achieve. Much recent work has sought to restore agreement among observers through institutions, gossip, empathy or synchronized opinions[17,18,20–22]. Agreement is necessary but not sufficient. Under one-step inertia, Shunning produced nearly unanimous agreement that everyone is bad. Evaluating a reputation mechanism therefore requires measures of both consensus and discrimination. Directional inertia improved both without any exchange of information between observers, because isolated errors are filtered within each observer's own records. It is thus complementary to mechanisms that synchronize opinions, and combining the two could further stabilize cooperation.

A related implication concerns the value of information. Quantitative scores and the aggregation of several observations show that using more than the latest observation can stabilize cooperation[28,29,32]. Our results add that the benefit of retained evidence depends on how it is allocated and saturates quickly. Allocations with the same depth produced very different outcomes, and under Scoring additional warning states beyond a few did not raise cooperation. A plausible reason is that longer warning paths also delay the condemnation of genuine defectors. This observation is consistent with the finding that individuals need not evolve to use all available information[32]. Compact and appropriately directed architectures offer a mechanistic explanation of why limited memory can suffice.

The vulnerability to opportunism exposes a general tension in reputation systems. Any mechanism that forgives isolated negative evidence lowers the marginal cost of occasional defection. This tension has been noted for generous assessment[24] and aggregation[32], and we find it for inertial updating and quantitative scoring as well. Two consequences follow. First, stability against unconditional defectors is not a sufficient test of a reputation system. Mutants that defect only occasionally can invade when unconditional defectors cannot, so invasion analyses should consider continuous families of partial defection. Second, the tension can be resolved without abandoning tolerance. Because inertial updating separates how much evidence changes a reputation from how strongly incomplete evidence is expressed, warning states can carry partial consequences while established reputations remain protected from isolated errors.

This principle matches the institutional practices that motivated our model. Graduated sanctions meet first offences with mild responses that escalate with repetition[34,35], and a criminal record loses its predictive value only after years without further offences[36]. Our model offers an evolutionary rationale for such arrangements. Responding moderately to early signs of misconduct, while reserving a full change of standing for repeated evidence, can protect cooperation from both errors and exploitation. The model also yields testable predictions for human impression formation. People should treat an individual with a single recent lapse less favorably than one with a clean record, without regarding that individual as bad. The speed at which they revise good and bad impressions should also depend on the prevailing norm. Evidence that impressions of good and bad individuals are updated asymmetrically[41,42] is consistent with this view, and experiments could test it directly.

Several limitations point to future work. The population-level theory relies on a mean-field closure that treats the records of different observers as conditionally independent, and strategy evolution is evaluated at stationary reputations. Finite-population simulations support the reputational results, but simulations in which strategies and records change simultaneously would test the separation of timescales. All individuals share one architecture, and memory carries no explicit cost. Allowing depths and strengths to evolve, and accounting for the cognitive cost of memory, would reveal which architectures emerge without external design. Finally, we tested resistance against unconditional mutants and independent occasional defection. Strategies that time their defections to the structure of the record, or whose omissions are correlated in time, deserve separate analysis.

Reputation systems are often discussed in terms of which behaviors should be rewarded or punished. Our results suggest that the dynamics of evidence matter as much. How much evidence a society requires before it changes a reputation, and how it treats individuals while that evidence accumulates, can decide whether cooperation based on private reputations succeeds.

Methods

Donation game, strategies and private assessment

We consider an infinitely large, well-mixed population in which randomly chosen donors and recipients play the donation game. A donor either cooperates, paying a cost $c > 0$ to give the recipient a benefit $b > c$, or defects, which has no payoff consequences. Individuals use one of three strategies $I \in \{\mathrm{ALLC}, \mathrm{ALLD}, \mathrm{DISC}\}$ with frequencies f_I, $\sum_I f_I = 1$. ALLC intends to cooperate with every recipient, ALLD intends to defect against every recipient, and DISC intends to cooperate if and only if it regards the recipient as good. All individuals share one social norm and one reputation mechanism, specified by the depths k_1, k_2 and the strengths α, β. Strategy evolution changes only the frequencies f_I.

Reputations are private[12,13,18]. Every individual keeps its own record of every other individual. In each interaction, an observer drawn independently of donor and recipient assesses the donor. A DISC donor chooses its action using its own image of the recipient, whereas the observer judges the donor's realized action using the observer's own image of the recipient, so the two can disagree. We use second-order norms[8]. Cooperation with a recipient regarded as good is assessed positively and defection against such a recipient negatively. Towards a recipient regarded as bad, cooperation and defection are assessed positively with probabilities p and q, respectively. SC, SH, SJ and SS correspond to $(p,q) = (1,0)$, $(0,0)$, $(0,1)$ and $(1,1)$.

Intended cooperation fails and becomes defection with probability e_1, whereas intended defection is always implemented. After applying the norm, the observer records the opposite assessment with probability e_2. The two errors are independent, and benefits and costs accrue only for realized cooperation. Let $P^+_{\mathrm{A}|\mathrm{Y}}$ denote the probability that the recorded assessment is positive, given the donor's intended action $A \in \{\mathrm{C}, \mathrm{D}\}$ and the observer's image $Y \in \{\mathrm{G}, \mathrm{B}\}$ of the recipient. With $\varepsilon = (1-e_1)(1-e_2) + e_1 e_2$,

$$\begin{aligned} P^+_{\mathrm{C}|\mathrm{G}} &= \varepsilon, & P^+_{\mathrm{C}|\mathrm{B}} &= e_2 + (1-2e_2)\big[(1-e_1)p + e_1 q\big], \\ P^+_{\mathrm{D}|\mathrm{G}} &= e_2, & P^+_{\mathrm{D}|\mathrm{B}} &= e_2 + (1-2e_2)q \end{aligned} \tag{5}$$

(Supplementary Note 1.1). Implementation error is already averaged into Eq. (5). If the observer regards the recipient as good with probability r, intended cooperation and intended defection are therefore assessed positively with probabilities

$$\Phi_{\mathrm{C}}(r) = rP^+_{\mathrm{C}|\mathrm{G}} + (1-r)P^+_{\mathrm{C}|\mathrm{B}}, \qquad \Phi_{\mathrm{D}}(r) = rP^+_{\mathrm{D}|\mathrm{G}} + (1-r)P^+_{\mathrm{D}|\mathrm{B}}, \tag{6}$$

and their difference, the assessment contrast, is $\Delta\Phi(r) = \Phi_{\mathrm{C}}(r) - \Phi_{\mathrm{D}}(r) = \kappa[r + (p-q)(1-r)]$ with $\kappa = (1-e_1)(1-2e_2)$. Unless stated otherwise, $b = 5$, $c = 1$ and $e_1 = e_2 = 0.02$.

Inertial reputation records and their interpretation

Observer i holds a latent record L_{ij} of target j in the state space

$$\mathscr{L} = \{\mathrm{G}, \mathrm{W}_1, \ldots, \mathrm{W}_{k_1}, \mathrm{B}, \mathrm{S}_1, \ldots, \mathrm{S}_{k_2}\}, \qquad k_1, k_2 \in \{0, 1, 2, \ldots\}. \tag{7}$$

After each interaction, only the observer's record of the donor changes, and only through the recorded assessment. The transitions follow Fig. 1b. A negative assessment moves $\mathrm{G} \to \mathrm{W}_1 \to \cdots \to \mathrm{W}_{k_1} \to \mathrm{B}$ and leaves B unchanged. A positive assessment moves $\mathrm{B} \to \mathrm{S}_1 \to \cdots \to \mathrm{S}_{k_2} \to \mathrm{G}$ and leaves G unchanged. A positive assessment returns every warning state W_m directly to G, and a negative assessment returns every recovery state S_n directly to B (direct reset). If $k_1 = 0$, a negative assessment moves G directly to B, and if $k_2 = 0$, a positive assessment moves B directly to G. Consecutive assessments are those made by the same observer about the same target. A record has $K+2$ states (Supplementary Note 1.2 and Supplementary Table S3).

Actions and assessments use a binary image that is sampled from the record whenever the record is consulted. The image is good with probability

$$\sigma(\mathrm{G}) = 1, \qquad \sigma(\mathrm{W}_m) = \alpha^m, \qquad \sigma(\mathrm{B}) = 0, \qquad \sigma(\mathrm{S}_n) = 1 - \beta^n, \qquad 0 \le \alpha, \beta \le 1. \tag{8}$$

Consultations are independent given the current state and do not change it. Setting $\alpha = \beta = 0$ at any depths also yields instantaneous updating, because the image then always equals the last recorded assessment (Supplementary Note 1.2). At one-step depths, $(\alpha, \beta) = (1,0)$ and $(0,1)$ give tolerance only and strictness only. The selected multi-step architectures are obtained from simulated reputation statistics, as described below.

Stationary reputation map

Consider one record whose successive recorded assessments are independent and positive with probability u. Its state then follows a Markov chain with transition matrix $T(u) = uT^+ + (1-u)T^-$, where T^+ and T^- encode the deterministic updates

after positive and negative assessments. For $0 < u < 1$ the chain is irreducible and aperiodic. Write $v = 1 - u$. Because W_m can be entered only from W_{m-1} (with $\mathrm{W}_0 \equiv \mathrm{G}$) after a negative assessment, and S_n only from S_{n-1} (with $\mathrm{S}_0 \equiv \mathrm{B}$) after a positive one, the stationary distribution satisfies $\pi_{\mathrm{W}_m} = v^m \pi_{\mathrm{G}}$ and $\pi_{\mathrm{S}_n} = u^n \pi_{\mathrm{B}}$. The only transitions from the good-standing states $\{\mathrm{G}, \mathrm{W}_m\}$ to the bad-standing states $\{\mathrm{B}, \mathrm{S}_n\}$ complete a run of $k_1 + 1$ negative assessments, and the only reverse transitions complete a run of $k_2 + 1$ positive assessments. Balancing these two fluxes gives $v^{k_1+1}\pi_{\mathrm{G}} = u^{k_2+1}\pi_{\mathrm{B}}$, and normalization yields

$$\pi_{\mathrm{G}} = \frac{u^{k_2+1}}{Z(u)}, \quad \pi_{\mathrm{W}_m} = \frac{v^m u^{k_2+1}}{Z(u)}, \quad \pi_{\mathrm{B}} = \frac{v^{k_1+1}}{Z(u)}, \quad \pi_{\mathrm{S}_n} = \frac{u^n v^{k_1+1}}{Z(u)}, \quad Z(u) = v^{k_1} + u^{k_2} - v^{k_1} u^{k_2} \tag{9}$$

(Supplementary Note 1.3). Averaging the image probabilities over this distribution defines the reputation map, the stationary probability that the target is regarded as good when it is assessed positively with probability u,

$$\mathscr{R}(u) \equiv \mathscr{R}(u; \alpha, \beta, k_1, k_2) = \sum_{\ell \in \mathscr{L}} \pi_\ell(u)\, \sigma(\ell) = \frac{u^{k_2+1} \sum_{m=0}^{k_1} (\alpha v)^m + v^{k_1+1} \sum_{n=1}^{k_2} u^n \left(1 - \beta^n\right)}{v^{k_1} + u^{k_2} - v^{k_1} u^{k_2}}. \tag{10}$$

Empty sums are zero, and $\mathscr{R}(0) = 0$ and $\mathscr{R}(1) = 1$ by continuity. Without inertia, $\mathscr{R}(u) = u$. For one-step inertia, $\mathscr{R}(u) = \{u^2[1 + \alpha(1-u)] + (1-\beta)u(1-u)^2\}/(1 - u + u^2)$, and with $\alpha = \beta = 1$, $\mathscr{R}(u) = u^{k_2}(1 - v^{k_1+1})/Z(u)$. The map increases with α and decreases with β. Two further properties are used below. For $k_1 \geq 1$, $\mathscr{R}'(1) = 1 - \alpha$, where the prime denotes $\mathrm{d}/\mathrm{d}u$, and for $k_2 = 0$, $\mathscr{R}$ is concave on $[0, 1]$ (Supplementary Note 1.3).

Self-consistent private reputations

Let r_I be the probability that a randomly chosen observer regards a randomly chosen type-I individual as good, and $\bar{r} = \sum_I f_I r_I$. We close the private-assessment problem with two mean-field assumptions, as in previous analyses of private assessment[13,18,21,32,39]. First, the donor's and the observer's images of a recipient are independent given the recipient's strategy. Both then regard a random recipient as good with probability $h_{\mathrm{GG}} = \sum_J f_J r_J^2$, both regard it as bad with probability $h_{\mathrm{BB}} = \sum_J f_J (1 - r_J)^2$, and each of the two ordered disagreements occurs with probability $h_{\mathrm{GB}} = \sum_J f_J r_J (1 - r_J)$. A type-$I$ donor is therefore assessed positively with probability

$$\begin{aligned} u_{\mathrm{ALLC}} &= \Phi_{\mathrm{C}}(\bar{r}), \qquad u_{\mathrm{ALLD}} = \Phi_{\mathrm{D}}(\bar{r}), \\ u_{\mathrm{DISC}} &= h_{\mathrm{GG}} P^+_{\mathrm{C|G}} + h_{\mathrm{GB}} \left(P^+_{\mathrm{C|B}} + P^+_{\mathrm{D|G}}\right) + h_{\mathrm{BB}} P^+_{\mathrm{D|B}}. \end{aligned} \tag{11}$$

Second, the assessments entering a record of a type-I individual are treated as independent with this common probability u_I, so that each record has the stationary response of Eq. (10). Stationary reputations then satisfy the self-consistency equations

$$r_I^* = \mathscr{R}\left(u_I(\boldsymbol{r}^*; \boldsymbol{f})\right), \qquad I \in \{\mathrm{ALLC}, \mathrm{ALLD}, \mathrm{DISC}\}. \tag{12}$$

Equation (12) can have several solutions. Consistent with records that start in good standing, we select at every composition the solution reached by the auxiliary relaxation

$$\frac{\mathrm{d}r_I}{\mathrm{d}\tau} = \mathscr{R}(u_I) - r_I, \qquad r_I(0) = 1. \tag{13}$$

The variable τ is a computational time, unrelated to evolutionary time t. The fixed points of Eq. (13) coincide with the stationary states of the mean-field dynamics of the full distribution of latent states, $\mathrm{d}\boldsymbol{\pi}_I/\mathrm{d}t' = \boldsymbol{\pi}_I[T(u_I) - \mathbf{I}]$ with $r_I = \boldsymbol{\pi}_I \boldsymbol{\sigma}$ (Supplementary Note 1.4). Stability statements refer to Eq. (13). Where checked, they agreed with the linearization of the full latent-state system.

In a homogeneous DISC population with reputation r, a resident donor regards its recipient as good with probability r, independently of the observer, and Eq. (11) reduces to

$$u_{\mathrm{R}}(r) = r\,\Phi_{\mathrm{C}}(r) + (1 - r)\,\Phi_{\mathrm{D}}(r). \tag{14}$$

Resident reputations are the roots of $F(r) \equiv \mathscr{R}[u_{\mathrm{R}}(r)] - r$, and their stability in Eq. (13) is given by the sign of $F'(r)$. A fold of a resident branch satisfies $F(r) = F'(r) = 0$, that is, $\mathscr{R}'[u_{\mathrm{R}}(r)]\, u'_{\mathrm{R}}(r) = 1$. We denote by r^*_{R} the highest strictly stable root, which coincides with the root reached by relaxation from $r = 1$ unless the highest root is a fold. We write $u^*_{\mathrm{R}} = u_{\mathrm{R}}(r^*_{\mathrm{R}})$, $\Phi^*_A = \Phi_A(r^*_{\mathrm{R}})$ and $\Delta\Phi^* = \Delta\Phi(r^*_{\mathrm{R}})$. Resident reputations do not depend on b/c.

Strategy evolution and cooperation

Reputations are assumed to equilibrate quickly relative to strategy change. At each strategy composition, payoffs are evaluated with the selected stationary reputations. Per unit time, in which each individual acts on average once as donor and once as recipient, the expected payoffs are

$$\begin{aligned}\Pi_{\mathrm{ALLC}} &= (1-e_1)\left[b\,(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}}\, r^*_{\mathrm{ALLC}}) - c\right],\\ \Pi_{\mathrm{ALLD}} &= (1-e_1)\,b\,(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}}\, r^*_{\mathrm{ALLD}}),\\ \Pi_{\mathrm{DISC}} &= (1-e_1)\left[b\,(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}}\, r^*_{\mathrm{DISC}}) - c\,\bar{r}^*\right],\end{aligned} \tag{15}$$

and strategy frequencies follow the replicator dynamics[37]

$$\frac{\mathrm{d}f_I}{\mathrm{d}t} = f_I\big(\Pi_I - \bar{\Pi}\big), \qquad \bar{\Pi} = \sum_J f_J \Pi_J. \tag{16}$$

The intended and realized cooperation rates are $\rho_{\mathrm{int}} = f_{\mathrm{ALLC}} + f_{\mathrm{DISC}}\,\bar{r}^*$ and $\rho = (1-e_1)\rho_{\mathrm{int}}$, and $\bar{\Pi} = (b-c)\rho$. In a homogeneous DISC population, $\rho_{\mathrm{int}} = r^*_{\mathrm{R}}$, and realized cooperation cannot exceed $1-e_1 = 0.98$.

To compare parameter settings globally, we integrate Eq. (16) from $n_{\mathrm{init}} = 512$ initial compositions sampled uniformly by area from the simplex, using the same compositions for all settings that are compared. In every scan, each trajectory ended either in the zero-cooperation set or at an attractor with positive cooperation. The zero-cooperation set comprises the ALLD vertex together with the low-reputation segment of the ALLD–DISC edge, along which the dynamics drift slowly towards ALLD (final $\rho \le 7\times10^{-4}$). Attractors with positive cooperation have final $\rho \ge 0.34$, and 0.974–0.980 in nearly all conditions. We therefore define the cooperative basin as the set of initial compositions whose trajectories reach an attractor with positive cooperation, and classify a trajectory as cooperative if its final $\rho \ge 0.01$. Any threshold inside the observed gap gives the same classification, and the gap was checked for every condition. We report the mean long-run cooperation $\bar{\rho}$ (the average of the final ρ over initial compositions), the cooperative-basin size ω_{C} (the fraction of cooperative trajectories) and the cooperation level $\bar{\rho}_{\mathrm{C}}$ within the basin, with $\bar{\rho} \approx \omega_{\mathrm{C}}\bar{\rho}_{\mathrm{C}}$. For the total-depth comparison (Fig. 4), the envelope at total depth K is the largest $\bar{\rho}$ among the $K+1$ allocations with $k_1 + k_2 = K$.

Invasion of a discriminator population

We consider a homogeneous DISC population at r^*_{R} and a rare mutant X that uses the residents' norm and record mechanism. Because the mutant is rare, it interacts with residents only and leaves their reputations and assessments unchanged. Suppose that X intends to help a resident recipient with probability χ_{X}, independently of the observer's image of that recipient, and acts independently across interactions. It is then assessed positively with probability $u_{\mathrm{X}} = \chi_{\mathrm{X}}\Phi^*_{\mathrm{C}} + (1-\chi_{\mathrm{X}})\Phi^*_{\mathrm{D}}$, its stationary reputation is $r^*_{\mathrm{X}} = \mathscr{R}(u_{\mathrm{X}})$, and its normalized payoff advantage over residents is (Supplementary Note 2.1)

$$\Lambda_{\mathrm{X}} \equiv \frac{\Pi_{\mathrm{X}} - \Pi_{\mathrm{R}}}{c(1-e_1)} = \big(r^*_{\mathrm{R}} - \chi_{\mathrm{X}}\big) - \frac{b}{c}\big(r^*_{\mathrm{R}} - r^*_{\mathrm{X}}\big), \tag{17}$$

the cooperation cost saved minus the benefit lost through reputation. The resident itself corresponds to $\chi = r^*_{\mathrm{R}}$ and $u = u^*_{\mathrm{R}} = r^*_{\mathrm{R}}\Phi^*_{\mathrm{C}} + (1-r^*_{\mathrm{R}})\Phi^*_{\mathrm{D}}$.

Unconditional mutants. ALLD ($\chi = 0$) has reputation $\mathscr{R}(\Phi^*_{\mathrm{D}})$ and ALLC ($\chi = 1$) has reputation $\mathscr{R}(\Phi^*_{\mathrm{C}})$, where $\Phi^*_{\mathrm{D}} = r^*_{\mathrm{R}}P^+_{\mathrm{D|G}} + (1-r^*_{\mathrm{R}})P^+_{\mathrm{D|B}}$ and Φ^*_{C} is defined analogously. When ALLD is regarded worse and ALLC better than the residents, Eq. (17) implies that the discriminator population resists both mutants if and only if b/c lies in the window of Eq. (1). If ALLD is not regarded worse than the residents, no b/c excludes it. If ALLC is not regarded better, it cannot invade. Because $u^*_{\mathrm{R}} = r^*_{\mathrm{R}}\Phi^*_{\mathrm{C}} + (1-r^*_{\mathrm{R}})\Phi^*_{\mathrm{D}}$, the window is non-empty if and only if $r^*_{\mathrm{R}}\mathscr{R}(\Phi^*_{\mathrm{C}}) + (1-r^*_{\mathrm{R}})\mathscr{R}(\Phi^*_{\mathrm{D}}) < \mathscr{R}(u^*_{\mathrm{R}})$, that is, if $\mathscr{R}$ at the resident's assessment probability lies above the chord joining its values at Φ^*_{D} and Φ^*_{C}. Without inertia, $\mathscr{R}$ is linear and $R_{\mathrm{L}} = R_{\mathrm{U}} = 1/\Delta\Phi^*$, so no benefit-to-cost ratio protects a discriminator population (Supplementary Note 2.2). Numerically, we classify local outcomes from the signs of the unnormalized payoff differences $\lambda_{\mathrm{ALLD}} = \Pi_{\mathrm{ALLD}} - \Pi_{\mathrm{R}}$ and $\lambda_{\mathrm{ALLC}} = \Pi_{\mathrm{ALLC}} - \Pi_{\mathrm{R}}$ (with $c = 1$), which also covers cases without finite thresholds. ALLD is not excluded if $\lambda_{\mathrm{ALLD}} \ge -10^{-10}$. Otherwise, ALLC invades if $\lambda_{\mathrm{ALLC}} > 10^{-10}$, and the residents resist both mutants if neither condition holds.

Opportunistic discriminators. The opportunistic discriminator (O-DISC) defined in the Results replaces each intended cooperation, independently and before implementation error, by intended defection with probability $\delta \in [0,1]$. Thus $\chi = (1-\delta)r^*_{\mathrm{R}}$. Its positive-assessment probability and reputation are

$$u_{\mathrm{X}}(\delta) = u^*_{\mathrm{R}} - \delta\, r^*_{\mathrm{R}}\,\Delta\Phi^*, \qquad r^*_{\mathrm{X}}(\delta) = \mathscr{R}\big[u_{\mathrm{X}}(\delta)\big], \tag{18}$$

and Eq. (17) gives the normalized invasion payoff of Eq. (2) and its slope at the resident, Eq. (3). By construction, $\Lambda(0) = 0$, and $\Lambda(1)$ is the normalized advantage of ALLD. For $r^*_{\mathrm{R}} > 0$, $\Lambda(\delta) = \delta r^*_{\mathrm{R}}[1 - (b/c)H(\delta)]$, where $H(\delta) = [r^*_{\mathrm{R}} - r^*_{\mathrm{X}}(\delta)]/(\delta r^*_{\mathrm{R}})$

is the reputation lost per unit of withheld cooperation, extended continuously by $H(0)=\Delta\Phi^{*}\mathscr{R}'(u_{\mathrm{R}}^{*})$. The residents therefore exclude every O-DISC mutant if and only if Eq. (4) holds, with equality permitting neutral mutants. Because $H(\delta)$ equals $\Delta\Phi^{*}$ times the slope of the chord of $\mathscr{R}$ between $u_{\mathrm{X}}(\delta)$ and u_{R}^{*}, for $\Delta\Phi^{*}>0$ the minimum is attained at $\delta\to 0$ whenever $\mathscr{R}$ is concave on that interval, which always holds for $k_2=0$. Equation (4) then reduces to $\Lambda'(0)\le 0$. In general, the whole interval must be checked. Near an all-good resident with $k_1\ge 1$, $\mathscr{R}'(u_{\mathrm{R}}^{*})\approx 1-\alpha$, so full tolerance strength makes weak opportunism cheap. In the error-free limit with $\beta=1$ and $k_1\ge 1$, an all-good resident excludes every O-DISC mutant if and only if $\alpha\le 1-c/b$, for every norm and every k_2 (Supplementary Note 2.3). At $b/c=5$ this bound equals the tolerance strength $\alpha=0.8$ used in Fig. 5 and Extended Data Fig. 9.

For fixed norm, depths, β and errors, a tolerance strength α is secure if a strictly stable high-reputation resident exists and satisfies Eq. (4). The lower boundary α_{L} of a secure interval, when it is not $\alpha=0$, is a fold of the high-reputation branch, $F(r,\alpha)=\partial_r F(r,\alpha)=0$ with $F(r,\alpha)=\mathscr{R}[u_{\mathrm{R}}(r);\alpha,\beta,k_1,k_2]-r$. The fold itself is excluded and does not depend on b/c. The upper boundary α_{U}, when it is not $\alpha=1$, is where Eq. (4) holds with equality. In every case we computed, it was determined by $\delta\to 0$, that is, by $(b/c)\,\Delta\Phi(r)\mathscr{R}'[u_{\mathrm{R}}(r);\alpha_{\mathrm{U}}]=1$ together with $F(r,\alpha_{\mathrm{U}})=0$. When the fold and the local condition meet, the secure interval closes. This happens at $b/c=u_{\mathrm{R}}'(r)/\Delta\Phi(r)$ evaluated at the fold, which equals 2 for Shunning (Supplementary Note 2.3).

Alternative reputation filters. For Fig. 5a we replace $\mathscr{R}$ by the stationary response of two alternative mechanisms while keeping the norm, errors, resident selection and Eqs. (2), (3) and (18) unchanged. Aggregation, adapted from Michel-Mata et al.[32], assigns a good image if at least Q of M independently sampled assessments are positive, $\mathscr{R}_{\mathrm{agg}}(u)=\sum_{j=Q}^{M}\binom{M}{j}u^{j}(1-u)^{M-j}$, with $(M,Q)=(2,1)$, so that $\mathscr{R}_{\mathrm{agg}}(u)=2u-u^{2}$. Quantitative assessment, adapted from Schmid et al.[28], keeps an integer score $z\in\{-z_{\max},\dots,z_{\max}\}$ that moves up or down by one after a positive or negative assessment and is interpreted as good if $z\ge z_0$. Attempts to leave the range leave the score unchanged. With $t=u/(1-u)$, $\mathscr{R}_{\mathrm{score}}(u)=\sum_{j=z_{\max}+z_0}^{2z_{\max}}t^{j}\big/\sum_{j=0}^{2z_{\max}}t^{j}$, and we use $z_{\max}=5$ and $z_0=0$. Neither mechanism was optimized or matched in memory to the inertial record (Supplementary Note 2.4).

Finite-population reputation simulations

Individual-based simulations test reputation formation without the mean-field closure, at fixed strategy composition. They do not simulate strategy evolution. A population of N individuals contains fixed numbers n_I of each strategy. Every DISC individual stores a latent record of every individual, including itself, and all records start in G. Records held by ALLC and ALLD individuals are not stored because they never affect actions or assessments. Each event consists of five steps. (i) A donor is drawn uniformly from the population and a recipient uniformly from the remaining $N-1$ individuals, and an observer is drawn independently and uniformly from the DISC individuals. The observer may coincide with the donor or the recipient. (ii) The donor forms its intended action, a DISC donor by sampling its own image of the recipient from Eq. (8). (iii) Implementation error is applied. (iv) The observer independently samples its own image of the recipient, applies the norm to the realized action and reverses the assessment with probability e_2. (v) The observer's record of the donor is updated. A sweep consists of $n_{\mathrm{DISC}}N$ events, one update per stored record on average.

At sampled times we use $s_{ij}=\sigma(L_{ij})$, which averages exactly over the randomness of image sampling. For target j, the mean goodness and the probability that two distinct DISC observers give different images are

$$g_j=\frac{1}{n_{\mathrm{DISC}}}\sum_{i}s_{ij},\qquad d_j=\binom{n_{\mathrm{DISC}}}{2}^{-1}\sum_{i<i'}\left[s_{ij}(1-s_{i'j})+(1-s_{ij})s_{i'j}\right],\tag{19}$$

where i and i' run over DISC observers. Averaging over type-I targets gives g_I and d_I, which are then averaged over sampling times. The reputational separation and the mean disagreement used in Fig. 2 and Extended Data Fig. 1 are

$$\Delta g=\frac{g_{\mathrm{DISC}}+g_{\mathrm{ALLC}}}{2}-g_{\mathrm{ALLD}},\qquad \bar{d}=\sum_{I}\frac{n_I}{N}d_I.\tag{20}$$

Unlike the mean-field theory, the simulations retain target-specific variation and correlations between the private records of different observers.

The recognition analyses use equal strategy numbers, $\alpha=\beta=1$ and $0\le k_1,k_2\le 6$. Each run lasts 1,000 sweeps and is sampled every five sweeps from sweep 800 to sweep 1,000 (41 samples). The depth maps in Fig. 2e–h use one run per condition with $N=150$. The replicated analysis (Extended Data Fig. 1) uses 30 independent runs per condition with $N=150$ and reports means across runs, with pointwise 95% percentile intervals from 2,000 bootstrap resamples of the run means. A condition is eligible for selection if, for both Δg and $\bar{d}$, the bootstrap interval of the difference between the second- and first-half means of the sampling window lies within $(-0.01,0.01)$. Two slowly relaxing SJ conditions required longer runs, $(k_1,k_2)=(4,5)$ with 3,000 sweeps (sampled from sweep 2,400) and $(3,5)$ with 50,000 sweeps (sampled from sweep 40,000). All 196 conditions then passed. Among eligible conditions with $\Delta g\ge 0.8$ and $\bar{d}\le 0.1$, we select the smallest K, breaking ties by larger Δg, then

smaller $\bar{d}$, k_1 and k_2. This gives the four selected architectures. The stars in Fig. 2e–h apply the same cutoffs and tie-breaking rule to the single-run maps, without the drift criterion, and identify the same four architectures. Selection uses reputation statistics only. The robustness analysis varies the separation cutoff from 0.70 to 0.90 in steps of 0.05 and the disagreement cutoff from 0.050 to 0.150 in steps of 0.025. The illustrative image matrices in Fig. 2a–d are final states of single runs with $N = 90$ after 1,000 sweeps.

Numerical procedures and figure-specific settings

All panels use $b/c = 5$ and $e_1 = e_2 = 0.02$ unless an axis varies them, and Supplementary Note 4 gives further numerical details. $\mathscr{R}$ and $\mathscr{R}'$ are evaluated in closed form. Homogeneous-DISC roots (Fig. 5 and Extended Data Figs. 2a–d, 3 and 9) are located by evaluating $F(r)$ on a fine uniform grid, refining sign changes by bracketing and classifying roots by the sign of $F'(r)$. At mixed compositions, Eq. (13) is integrated from $r_I = 1$ by fourth-order Runge–Kutta steps until $\max_I |\mathscr{R}(u_I) - r_I| \leq 10^{-11}$ (10^{-12} for Fig. 3). For local invasion (Extended Data Figs. 5–8), the resident relaxation is followed by step-limited Newton refinement, which keeps the branch reached from $r = 1$, to residual $\leq 10^{-13}$.

All sampled-cooperation results (Fig. 4 and Extended Data Figs. 2e–h, 5, 6e–h and 8) use one engine. Selected reputations are tabulated exactly on a triangular lattice with 120 subdivisions and interpolated linearly only inside cells in which no reputation component varies by more than 0.02. Elsewhere, including every cell crossed by a fold, they are computed exactly at the requested composition. Payoffs and the replicator field are always computed from these reputations, never interpolated. Trajectories are integrated with the explicit midpoint rule ($\Delta t = 0.2$) until ρ_{int} changes by at most 10^{-7} over each of three consecutive intervals of 100 time units, or for at most 4×10^4 time units. Halving the step, refining the lattice and tightening the tolerances changed $\bar{\rho}$ by at most 1.3×10^{-5} and changed no outcome class. Initial compositions are 512 points of a randomly shifted Halton sequence (bases 2 and 3, seed 2026) or, for Extended Data Fig. 5, eight independently scrambled 512-point Sobol sequences[43] (seeds 2026–2033). Points are mapped area-uniformly onto the simplex by reflecting points of the unit square with $x + y > 1$. In Extended Data Fig. 5, standard errors are computed across the eight samples and therefore describe the sampling of initial compositions. The dynamics themselves are deterministic. Extended Data Fig. 2e–h initializes reputations once, at $t = 0$, as all-good or all-bad and then follows them quasi-statically along each trajectory by warm-started relaxation, so that a branch is left only where it ends.

In Fig. 3 and Extended Data Fig. 6a–d, reputations are relaxed exactly at every point of an 84×74 grid clipped to the simplex. Vertex stability follows from the two rare-strategy growth rates. Edge equilibria are located on 501-point edge scans, and their markers refer to stability along the edge. Interior equilibria are classified by their Jacobian. Under SC without inertia, all three payoffs are equal on the line $f_{\mathrm{DISC}} = c/[b(\varepsilon - e_2)] \approx 0.2126$ (Supplementary Note 1.5). Extended Data Fig. 3 reports the highest stable homogeneous-DISC root and marks bistability (one-step (α, β) plane, 161×161; depth plane, $0 \leq k_1, k_2 \leq 8$). For every root in the depth planes, stability in Eq. (13) agreed with the linearization of the full latent-state dynamics. On the ALLD–DISC edge (Extended Data Fig. 4), Eq. (12) reduces exactly to one equation in $\bar{r}$, which is solved for 1,071 values of $f_{\mathrm{DISC}} \in [0.01, 1]$. Local invasion maps use 241×241 points of $e_1, e_2 \in [0, 0.45]$ (Extended Data Fig. 7) and 801×801 points of (α, β) (Extended Data Fig. 6i–l). In Fig. 5, $\Lambda(\delta)$ is evaluated at 5,015 values of δ, concentrated near $\delta = 0$, with sign changes refined by Brent's method. The endpoint δ_c is the upper end of the interval with $\Lambda > 0$. Residents of the alternative filters are obtained from $r = 1$ by damped iteration, $r \leftarrow 0.8r + 0.2\mathscr{R}_{\mathrm{X}}[u_{\mathrm{R}}(r)]$ with $\mathscr{R}_{\mathrm{X}} \in \{\mathscr{R}_{\mathrm{agg}}, \mathscr{R}_{\mathrm{score}}\}$. For Extended Data Fig. 9 (at least 601 values of α, 161 values of $b/c \in [2, 8]$ and $\beta = 1$), $\min_\delta H(\delta)$ is found by comparing H at $\delta = 0$, $\delta = 1$ and all real roots in $(10^{-8}, 1)$ of the numerator of the rational function $\mathrm{d}H/\mathrm{d}\delta$. A tolerance strength is classified as secure if $1 - (b/c)\min_\delta H \leq 10^{-10}$. Fold endpoints are excluded, whereas endpoints at the domain boundary or at neutral invasion are included.

Data availability

All numerical data underlying the figures are included in the verified computational package that accompanies the paper. [Repository and DOI to be added before submission.]

Code availability

The code and figure scripts used in this study are provided in the same package, with one notebook per figure, a C engine for the strategy dynamics, input configurations, validation records and checksums. [Repository and DOI to be added before submission.]

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[To be added.]

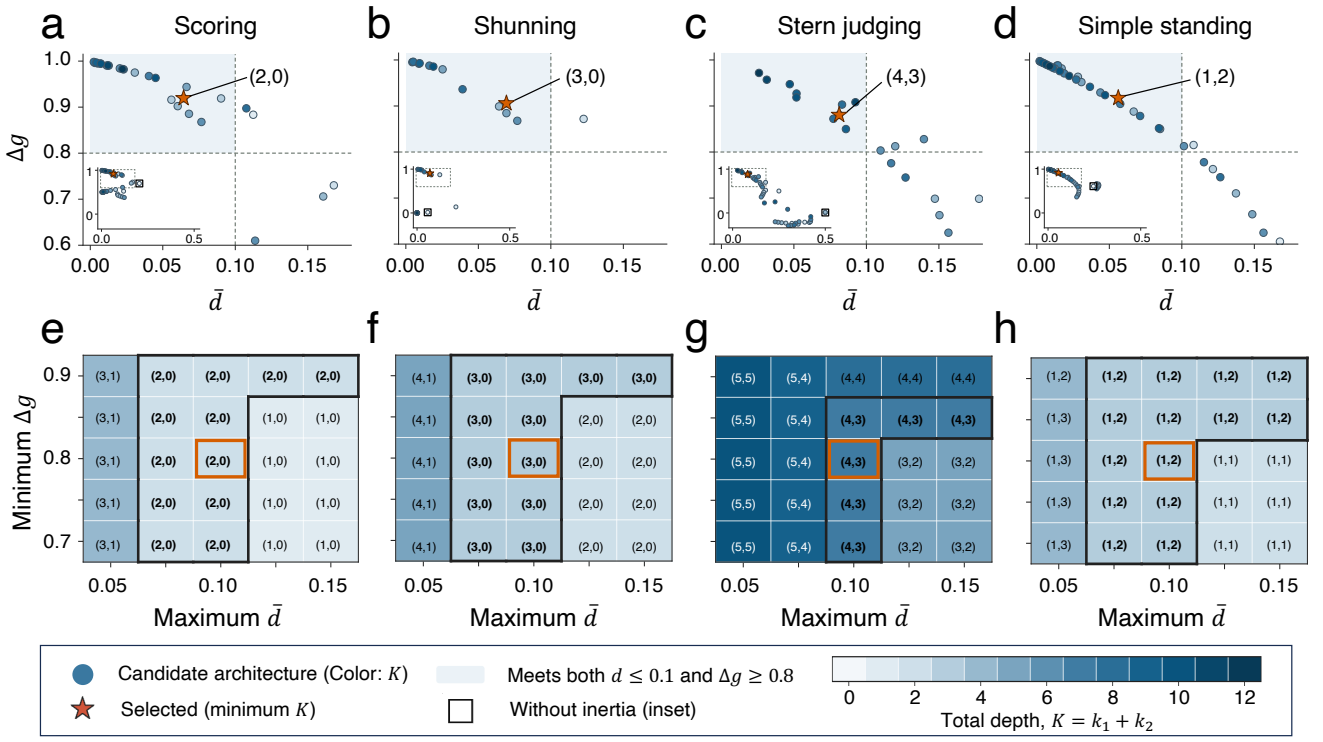
Author contributions

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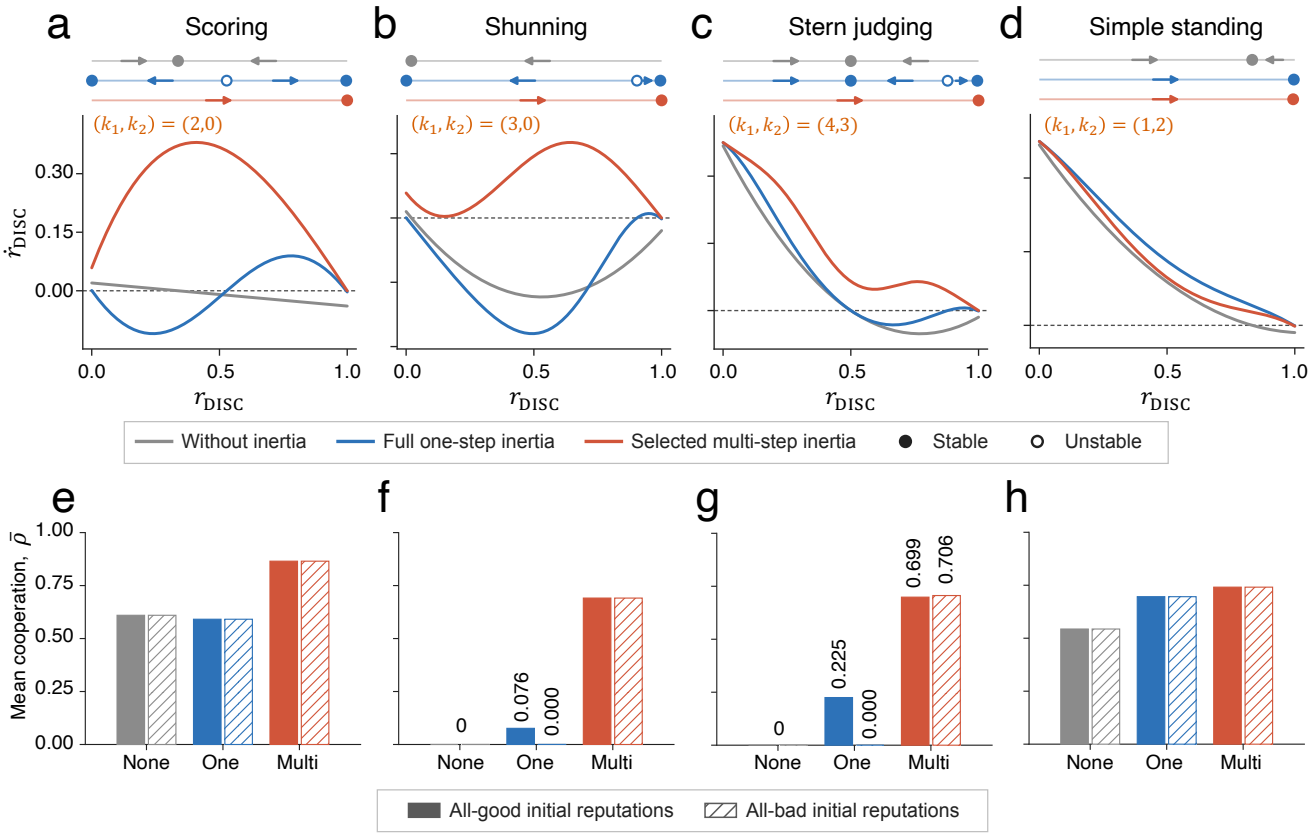
Competing interests

[To be added.]

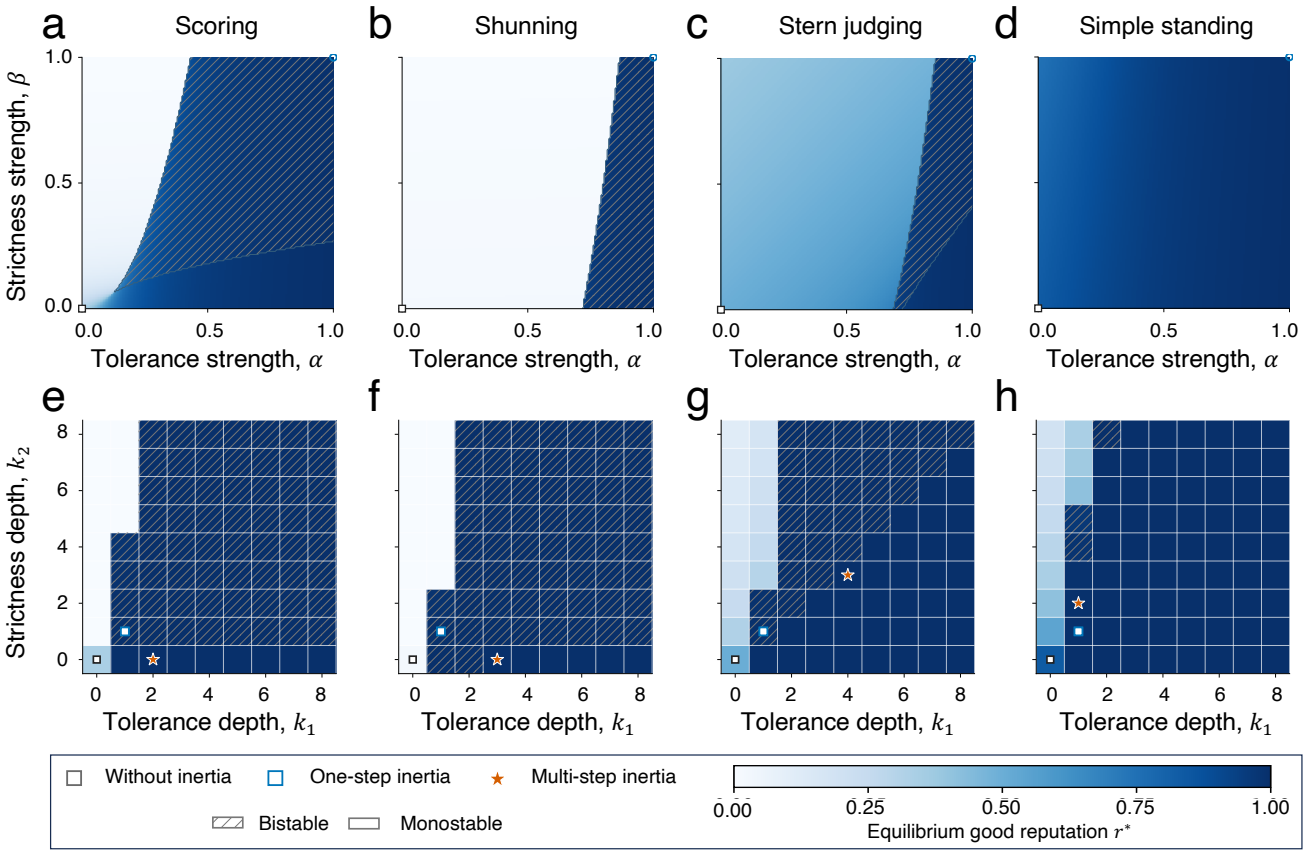
Extended Data



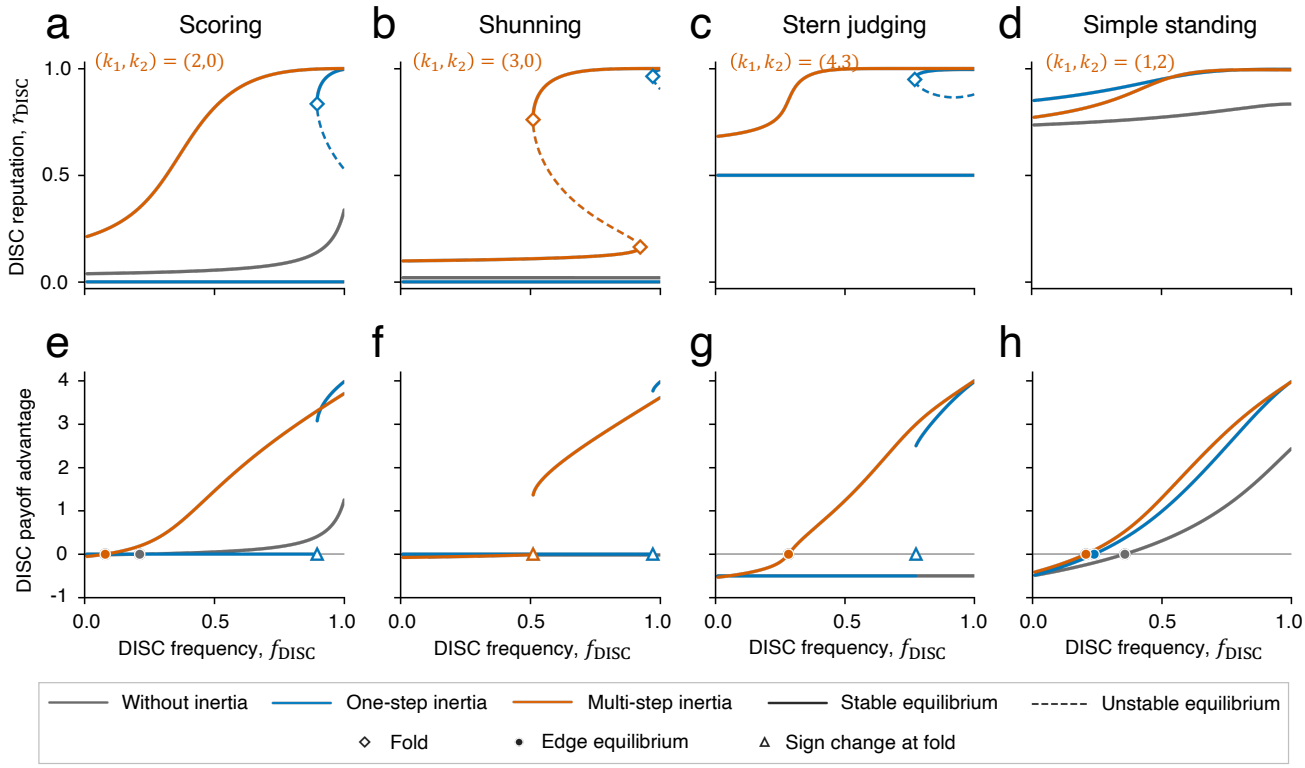
Extended Data Fig. 1. Selection and robustness of representative inertia architectures. a–d, Reputational separation Δg and mean observer disagreement $\bar{d}$ for the tested depth pairs under SC, SH, SJ and SS, respectively. Points are colored by total depth $K = k_1 + k_2$, and the shaded region marks the joint criteria $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$. Stars identify the qualifying pairs with the smallest K, namely $(2,0)$, $(3,0)$, $(4,3)$ and $(1,2)$, respectively. Insets display the wider parameter space, with open squares marking instantaneous updating. **e–h**, Minimum-depth pair selected when the minimum separation and the maximum disagreement are varied. Orange boxes mark the default criteria, and dark outlines mark the region that retains the default selection. Although the exact depth pairs depend on the thresholds, compact qualifying architectures persist over a range of criteria. Each estimate averages 30 independent simulations, and only conditions that pass the drift criterion are eligible (Methods). Parameters: equal proportions of the three strategies, $N = 150$, $e_1 = e_2 = 0.02$ and $\alpha = \beta = 1$.



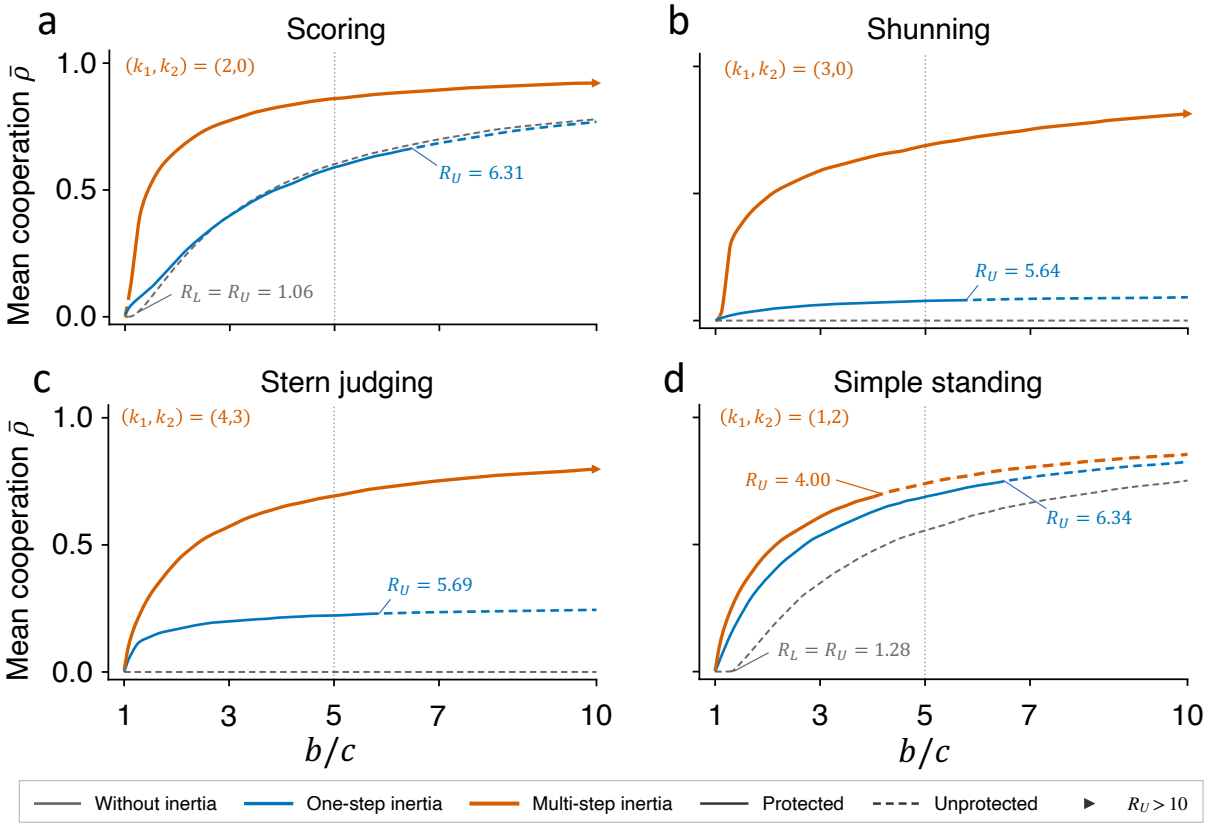
Extended Data Fig. 2. Reputation equilibria and population outcomes under contrasting initial reputations. a–d, Reputation dynamics in homogeneous DISC populations under SC, SH, SJ and SS, shown as the rate of change of the DISC good-reputation probability r_{DISC} in the auxiliary relaxation (Methods). Gray, blue and red curves represent instantaneous updating, full one-step inertia and the selected multi-step architectures, respectively. Filled and open circles denote stable and unstable equilibria, and arrows indicate the direction of the reputation dynamics. Full one-step inertia supports both high and low stable reputation states under SC, SH and SJ, whereas the selected multi-step architectures yield a single high-reputation equilibrium. **e–h**, Mean realized cooperation $\bar{\rho}$ across initial strategy compositions under SC, SH, SJ and SS after all-good (solid bars) and all-bad (hatched bars) initial reputations. The selected multi-step architectures produce similar mean cooperation under both initial conditions, with a small residual difference under SJ. Under one-step inertia, cooperation under SH and SJ collapses after all-bad initialization. Parameters: $b/c = 5$, $e_1 = e_2 = 0.02$ and $\alpha = \beta = 1$.



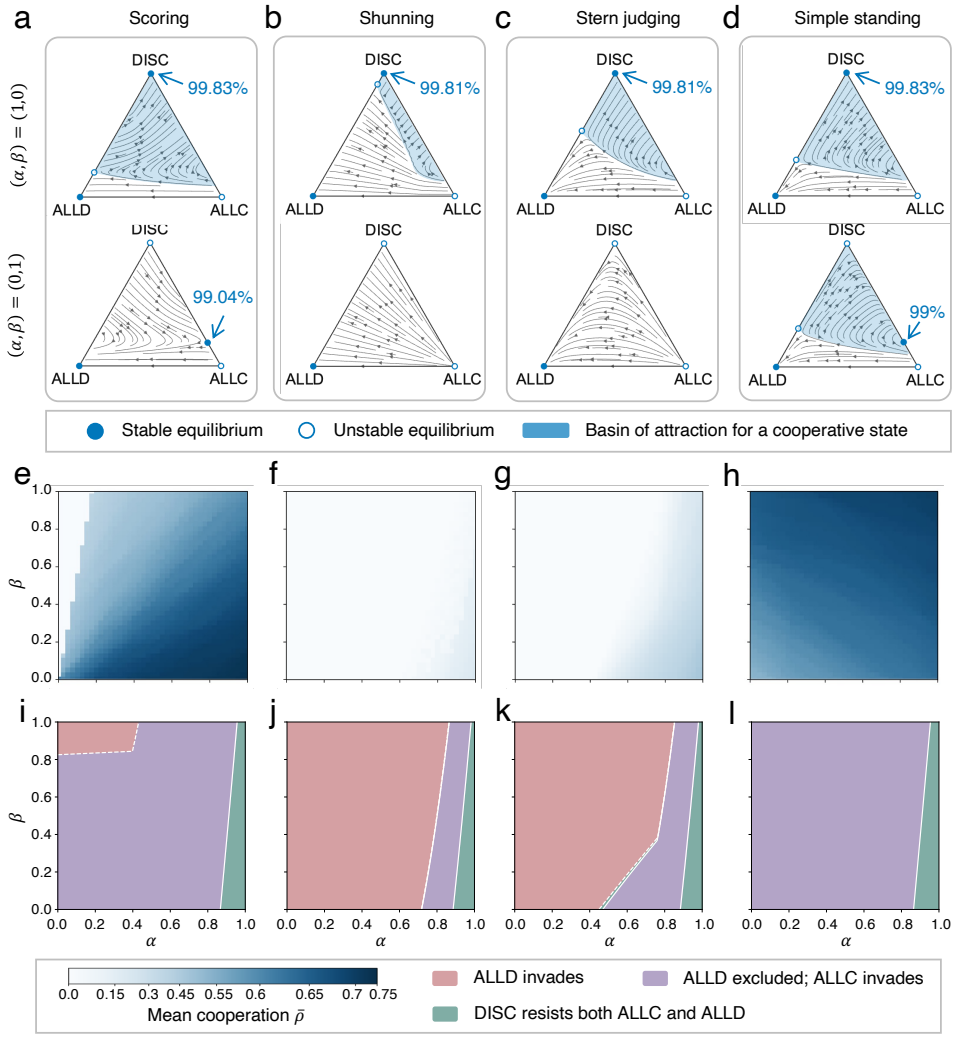
Extended Data Fig. 3. Tolerance, strictness and depth jointly determine reputation equilibria. a–d, Highest stable good-reputation equilibrium in a homogeneous DISC population across tolerance strength α and strictness strength β under one-step inertia ($k_1 = k_2 = 1$), for SC, SH, SJ and SS. **e–h**, Highest stable equilibrium across tolerance depth k_1 and strictness depth k_2 with $\alpha = \beta = 1$. Color denotes the equilibrium good-reputation probability, and hatching identifies bistability in the homogeneous-DISC reputation dynamics. Open and blue squares mark instantaneous updating and full one-step inertia, and stars mark the selected multi-step architectures. High-reputation equilibria and the presence of bistability depend on the norm and on the direction and strength of inertia. Parameters: $e_1 = e_2 = 0.02$.



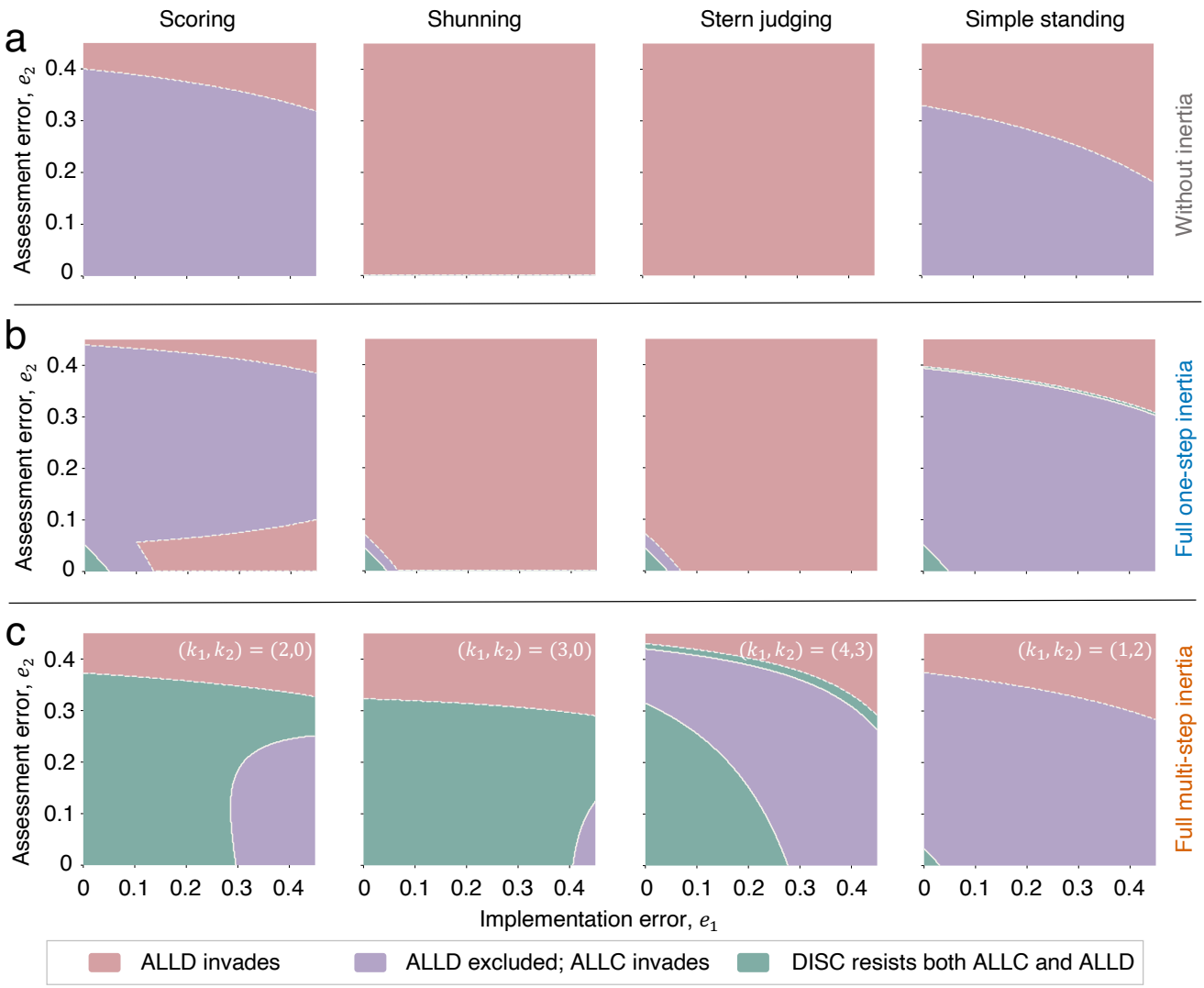
Extended Data Fig. 4. Reputation branches and payoff thresholds on the ALLD–DISC edge. a–d, Equilibrium good reputation of DISC, r_{DISC}, as its frequency f_{DISC} varies along the ALLD–DISC edge under SC, SH, SJ and SS. Gray, blue and orange show instantaneous updating, full one-step inertia and selected multi-step inertia. Solid and dashed curves are stable and unstable reputation equilibria, respectively, and open diamonds mark folds where branches meet. **e–h**, Corresponding normalized payoff advantage of DISC over ALLD on the branch selected by all-good initialization. Filled circles mark continuous edge equilibria, and triangles mark discontinuous changes in payoff sign at folds. Under one-step SC, SH and SJ, the high-reputation branch disappears when DISC becomes sufficiently rare, causing an abrupt loss of its payoff advantage. The selected multi-step architecture moves this threshold to lower DISC frequencies under SH and removes the fold under SC and SJ, where the threshold becomes a continuous edge equilibrium. Under SS, all thresholds are continuous. Parameters: $b/c = 5$, $e_1 = e_2 = 0.02$ and, for the inertial settings, $\alpha = \beta = 1$.



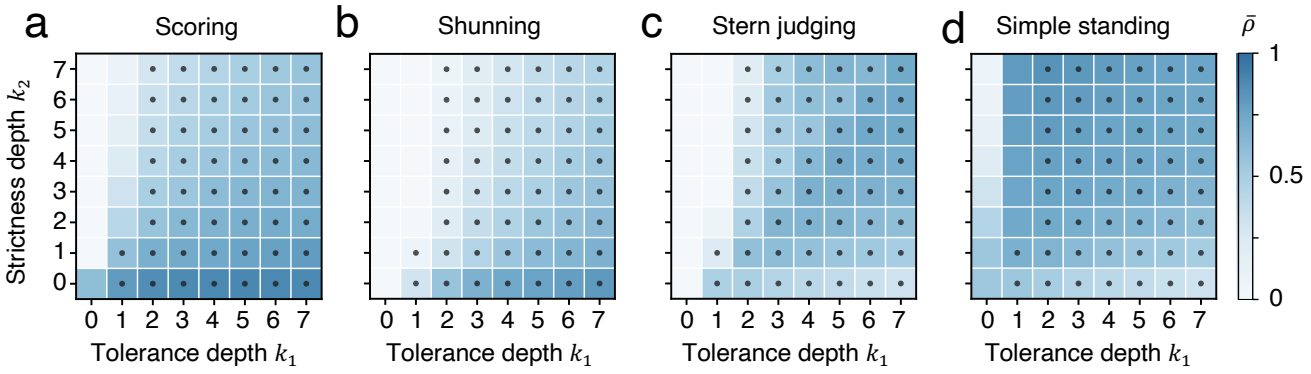
Extended Data Fig. 5. Cooperation and local discriminator protection across benefit-to-cost ratios. a–d, Mean cooperation $\bar{\rho}$ across initial strategy compositions as b/c varies under SC, SH, SJ and SS. Gray, blue and orange lines denote instantaneous updating, full one-step inertia and the selected multi-step architectures. Solid segments indicate local protection of pure DISC against ALLC and ALLD ($R_L < b/c < R_U$; Eq. (1)), whereas dashed segments indicate that at least one of these mutants can invade pure DISC. Labels give the window bounds, and arrowheads indicate $R_U > 10$. The vertical dotted line marks $b/c = 5$. Curves are means over eight independent samples of 512 initial compositions (Methods). Selected multi-step inertia supports higher mean cooperation across much of the range. Under SS, high cooperation can persist even where pure DISC is not locally protected. Parameters: $e_1 = e_2 = 0.02$ and $\alpha = \beta = 1$.



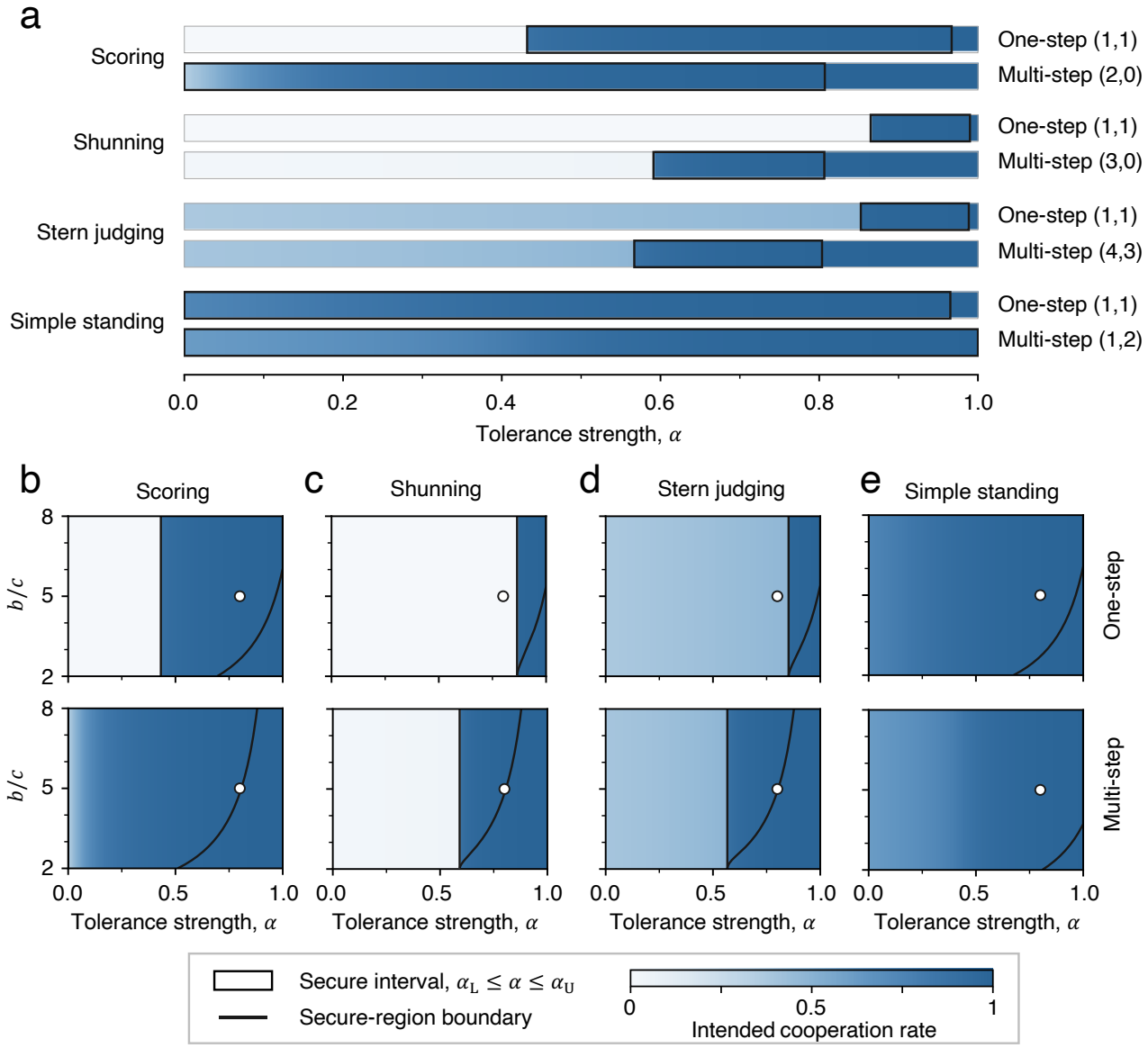
Extended Data Fig. 6. Separate effects of tolerance and strictness on strategy dynamics. a–d, Strategy dynamics under SC, SH, SJ and SS. In each column, the upper simplex shows tolerance only, $(\alpha,\beta) = (1,0)$, and the lower simplex shows strictness only, $(\alpha,\beta) = (0,1)$. Arrows indicate the direction of evolution, filled and open circles denote stable and unstable equilibria, shading marks the cooperative basin, and percentages indicate intended cooperation at the highlighted equilibria. **e–h**, Mean cooperation $\bar{\rho}$ across the one-step (α,β) plane. **i–l**, Local invasion classes for a pure-DISC resident. Pink indicates that ALLD invades, purple that ALLD is excluded but ALLC invades, and green that DISC resists both mutants. Tolerance and strictness have different, norm-dependent effects on strategy dynamics. Parameters: $b/c = 5$, $e_1 = e_2 = 0.02$ and $k_1 = k_2 = 1$.



Extended Data Fig. 7. Local invasion resistance across implementation and assessment errors. a–c, Local invasion classes of a pure-DISC resident across implementation error e_1 and assessment error e_2 under instantaneous updating (**a**), full one-step inertia (**b**) and selected multi-step inertia (**c**), with SC, SH, SJ and SS in columns. Pink denotes that ALLD invades, purple that ALLD is excluded but ALLC invades, and green that DISC resists both mutants. Without inertia or with one-step inertia, ALLD invades almost everywhere under SH and SJ. Selected multi-step inertia excludes ALLD over most of the error plane under these norms and, under SC and SH, makes DISC resistant to both mutants over most of it. These maps concern local resistance only and are not cooperation scans. Parameters: $b/c = 5$ and $\alpha = \beta = 1$.



Extended Data Fig. 8. Mean cooperation and local discriminator protection across inertia depths. a–d, Mean realized cooperation $\bar{\rho}$ over the (k_1, k_2) plane under SC, SH, SJ and SS, respectively. Color denotes cooperation averaged over initial strategy compositions, and dots identify depth pairs at which a pure-DISC population resists rare ALLC and ALLD mutants. SC and SH show high cooperation for tolerance-dominated allocations, SJ benefits from a combination of tolerance and strictness depths, and SS supports high cooperation whenever a warning path is present, with the highest values for strictness-weighted allocations. The dots also show that protection of pure DISC and high population-level cooperation are not identical. Parameters: $b/c = 5$, $e_1 = e_2 = 0.02$ and $\alpha = \beta = 1$.



Extended Data Fig. 9. Tolerance strength and benefit-to-cost ratio shape resistance to opportunistic defection. a, Secure intervals in which an established DISC population resists the tested O-DISC family at $b/c = 5$, comparing the one-step architecture $(k_1, k_2) = (1, 1)$ with the selected multi-step architecture for each norm. Blue shading gives resident intended cooperation. Black outlines identify numerically secure intervals, and interval endpoints that coincide with branch folds are excluded. **b–e**, Resident intended cooperation and O-DISC security across α and b/c under SC, SH, SJ and SS, respectively. The upper and lower rows show one-step and selected multi-step inertia. Black curves delimit the secure region, and white circles mark $(\alpha, b/c) = (0.8, 5)$. At this reference point the selected multi-step architectures maintain high resident cooperation and resist the tested opportunistic family under all four norms, whereas the outcome under one-step inertia depends on the norm. Security against O-DISC and high resident cooperation are distinct quantities. The selected multi-step depths are $(2, 0)$, $(3, 0)$, $(4, 3)$ and $(1, 2)$ for SC, SH, SJ and SS, respectively. Parameters: $e_1 = e_2 = 0.02$ and $\beta = 1$.

Supplementary Information

Indirect reciprocity with inertial reputation updating

Wenqiang Zhu et al.

This Supplementary Information provides the derivations and numerical details behind the Methods. Supplementary Note 1 develops the model and its theory, from a single interaction to the stationary reputation map, population reputations and strategy dynamics. Supplementary Note 2 analyzes invasion of a discriminator population by unconditional and opportunistic mutants. Supplementary Notes 3 and 4 describe the finite-population simulations and the numerical methods, and Supplementary Note 5 summarizes the notation. Equation numbers without the prefix S refer to the main text and Methods. [The Supplementary Information will be revised further.]

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Supplementary Note 1. Model and theoretical framework

Two notational conventions are used throughout. The letters G and B name both the two values of a binary image (good, bad) and the two established latent states. Because an established state is always interpreted as the image of the same name, this use is unambiguous. Subscripts C and D denote intended actions, strategy-level quantities carry full strategy names (r_{ALLC}, u_{DISC}, ...), the subscript R denotes the resident of a homogeneous DISC population and X a rare mutant. Supplementary Table S11 lists all symbols.

1.1 Interactions, social norms and errors

Sequence of an interaction. A donor, a recipient and an observer are drawn from the population, the observer independently of the two others. Each interaction proceeds as follows.

1. The donor forms an intended action. ALLC intends C, ALLD intends D, and DISC consults its own record of the recipient, samples a binary image and intends C if and only if the image is G.
2. Intended cooperation is implemented as defection with probability e_1, whereas intended defection is always implemented. The recipient receives b and the donor pays c only if cooperation is realized.
3. The observer consults its own record of the recipient, samples a binary image Y and applies the norm to the realized action. A positive assessment is prescribed with probability ν_{AY}, with $\nu_{\mathrm{CG}} = 1$, $\nu_{\mathrm{DG}} = 0$, $\nu_{\mathrm{CB}} = p$ and $\nu_{\mathrm{DB}} = q$.
4. The prescribed assessment is reversed with probability e_2.
5. The recorded assessment updates the observer's record of the donor. No other record changes.

The donor's and the observer's images of the recipient come from different records, so the donor may act on a different view of the recipient from the one used to judge it. This is the defining feature of private assessment.

Error-integrated assessment probabilities. Conditional on a realized action A and an image Y, the recorded assessment is positive with probability $\nu_{AY}(1-e_2)+(1-\nu_{AY})e_2 = e_2+(1-2e_2)\nu_{AY}$. Intended cooperation is realized as C with probability $1-e_1$ and as D with probability e_1, whereas intended defection is always realized as D. Averaging over the realized action,

$$P^+_{\mathrm{C}|\mathrm{Y}} = e_2+(1-2e_2)\big[(1-e_1)\nu_{\mathrm{CY}}+e_1\nu_{\mathrm{DY}}\big], \qquad P^+_{\mathrm{D}|\mathrm{Y}} = e_2+(1-2e_2)\nu_{\mathrm{DY}}, \tag{S1}$$

which gives Eq. (5) of the Methods. Writing $\kappa = (1-e_1)(1-2e_2) = \varepsilon - e_2$, the probability $P^+_{\mathrm{C}|\mathrm{B}}$ can also be written as $p(\varepsilon-e_2)+q(1-\varepsilon-e_2)+e_2$, because $1-\varepsilon-e_2 = e_1(1-2e_2)$. Supplementary Table S1 gives the four norms. These probabilities already include implementation error. The separate factor $1-e_1$ in the payoffs accounts for realized donations, not for assessments.

Table S1. Norms and error-integrated assessment probabilities.

Norm	$p=\nu_{\mathrm{CB}}$	$q=\nu_{\mathrm{DB}}$	$P^+_{\mathrm{C}\|\mathrm{G}}$	$P^+_{\mathrm{D}\|\mathrm{G}}$	$P^+_{\mathrm{C}\|\mathrm{B}}$	$P^+_{\mathrm{D}\|\mathrm{B}}$
Scoring (SC)	1	0	ε	e_2	ε	e_2
Shunning (SH)	0	0	ε	e_2	e_2	e_2
Stern Judging (SJ)	0	1	ε	e_2	$1-\varepsilon$	$1-e_2$
Simple Standing (SS)	1	1	ε	e_2	$1-e_2$	$1-e_2$

Action-conditional assessment probabilities. If the observer regards the recipient as good with probability r, independently of the donor's image, intended cooperation and intended defection are assessed positively with the probabilities $\Phi_{\mathrm{C}}(r)$ and $\Phi_{\mathrm{D}}(r)$ of Eq. (6). Both are linear in r, with slopes $\Phi'_{\mathrm{C}} = P^+_{\mathrm{C}|\mathrm{G}}-P^+_{\mathrm{C}|\mathrm{B}}$ and $\Phi'_{\mathrm{D}} = P^+_{\mathrm{D}|\mathrm{G}}-P^+_{\mathrm{D}|\mathrm{B}}$. Their difference is

$$\Delta\Phi(r) = r\big(P^+_{\mathrm{C}|\mathrm{G}}-P^+_{\mathrm{D}|\mathrm{G}}\big)+(1-r)\big(P^+_{\mathrm{C}|\mathrm{B}}-P^+_{\mathrm{D}|\mathrm{B}}\big) = \kappa\big[r+(p-q)(1-r)\big], \tag{S2}$$

because $P^+_{\mathrm{C}|\mathrm{G}}-P^+_{\mathrm{D}|\mathrm{G}} = \kappa$ and $P^+_{\mathrm{C}|\mathrm{B}}-P^+_{\mathrm{D}|\mathrm{B}} = (p-q)\kappa$. Thus $\Delta\Phi = \kappa$ (SC), κr (SH, SS) and $\kappa(2r-1)$ (SJ). Under SJ the contrast changes sign at $r=1/2$, so on a low-reputation branch defection is assessed better than cooperation.

A DISC donor in a homogeneous DISC population with reputation r regards the recipient as good with probability r, independently of the observer (Supplementary Note 1.4). It therefore intends C with probability r and is assessed positively with probability

$$u_{\mathrm{R}}(r) = r\Phi_{\mathrm{C}}(r)+(1-r)\Phi_{\mathrm{D}}(r) = r^2P^+_{\mathrm{C}|\mathrm{G}}+r(1-r)\big(P^+_{\mathrm{C}|\mathrm{B}}+P^+_{\mathrm{D}|\mathrm{G}}\big)+(1-r)^2P^+_{\mathrm{D}|\mathrm{B}}, \tag{S3}$$

with derivative $u'_{\mathrm{R}}(r) = \Delta\Phi(r)+r\Phi'_{\mathrm{C}}+(1-r)\Phi'_{\mathrm{D}}$. Supplementary Table S2 lists these functions.

Table S2. Resident assessment functions. $\kappa = (1-e_1)(1-2e_2)$. For $e_1=e_2=0.02$, $\varepsilon = 0.9608$ and $\kappa = 0.9408$.

Norm	$\Phi_{\mathrm{C}}(r)$	$\Phi_{\mathrm{D}}(r)$	$\Delta\Phi(r)$	$u_{\mathrm{R}}(r)$
SC	ε	e_2	κ	$e_2+\kappa r$
SH	$e_2+\kappa r$	e_2	κr	$e_2+\kappa r^2$
SJ	$1-\varepsilon-(1-2\varepsilon)r$	$1-e_2-(1-2e_2)r$	$\kappa(2r-1)$	$1-e_2-(1-2e_2+\kappa)r+2\kappa r^2$
SS	$1-e_2-(1-e_2-\varepsilon)r$	$1-e_2-(1-2e_2)r$	κr	$1-e_2-(1-2e_2)r+\kappa r^2$

1.2 The inertial reputation record

States and transitions. Each observer holds, for each target, a latent record in $\mathscr{L} = \{\mathrm{G},\mathrm{W}_1,\dots,\mathrm{W}_{k_1},\mathrm{B},\mathrm{S}_1,\dots,\mathrm{S}_{k_2}\}$ with $k_1,k_2\in\{0,1,2,\dots\}$. The established states G and B represent good and bad standing. Warning states W_m count consecutive negative assessments that challenge good standing, and recovery states S_n count consecutive positive assessments that challenge bad standing. Supplementary Table S3 lists the transitions. Two rules generate it. Evidence against the established judgment advances along the corresponding path, and evidence supporting it resets the whole path (direct reset). The record has $K+2$ states. Consecutiveness refers to the assessments made by one observer about one target, which need not be consecutive interactions of the target.

We order the states as $(\mathrm{G},\mathrm{W}_1,\dots,\mathrm{W}_{k_1},\mathrm{B},\mathrm{S}_1,\dots,\mathrm{S}_{k_2})$ and write T^+ and T^- for the $(K+2)\times(K+2)$ matrices with entry $[T^{\pm}]_{\ell\ell'} = 1$ if a positive (negative) assessment moves ℓ to ℓ' and 0 otherwise. Each row of $T^{\pm}$ contains exactly one 1. If a

Table S3. Transitions of a latent record. Rows for W_m and S_n are absent when $k_1 = 0$ or $k_2 = 0$, respectively.

Current state	After a positive assessment	After a negative assessment
G	G	W_1 if $k_1 \geq 1$; B if $k_1 = 0$
W_m, $1 \leq m < k_1$	G	W_{m+1}
W_{k_1}	G	B
B	S_1 if $k_2 \geq 1$; G if $k_2 = 0$	B
S_n, $1 \leq n < k_2$	S_{n+1}	B
S_{k_2}	G	B

record receives a positive assessment with probability u and a negative one with probability $v = 1 - u$, its state follows the Markov chain with transition matrix

$$T(u) = uT^{+} + vT^{-}, \qquad \text{for example} \qquad T(u) = \begin{pmatrix} u & v & 0 & 0 \\ u & 0 & v & 0 \\ 0 & 0 & v & u \\ u & 0 & v & 0 \end{pmatrix} \tag{S4}$$

for one-step inertia in the order (G, W_1, B, S_1).

Binary images. When a record in state ℓ is consulted, it yields the image G with probability $\sigma(\ell)$ given by Eq. (8), and B otherwise. Each consultation draws a fresh image, independently of all other consultations given the current state, and does not change the state. In particular, when a DISC donor is also the observer of its own action, the image used to act and the image used to judge are independent draws from the same record. The geometric form makes successive warning states progressively less likely to be treated as good, and successive recovery states progressively more likely to be treated as good. The depths control how much consistent evidence completes a reversal of the latent state, whereas the strengths control how strongly incomplete evidence already affects behavior and assessment.

Settings used in the paper. Supplementary Table S4 lists the reference parameters and the three inertia settings that are compared throughout. Instantaneous updating can be implemented in two equivalent ways.

Lemma S1. *If $(k_1, k_2) = (0, 0)$, or if $\alpha = \beta = 0$ at any depths, the image of a record that has received at least one assessment equals its most recent assessment, and a record that has received none yields G.*

Proof. For $(k_1, k_2) = (0, 0)$ the record has the states G and B only, and every assessment sets it to G or B according to its sign. For $\alpha = \beta = 0$ and any depths, the states that can follow a positive assessment are G and $S_1, \ldots, S_{k_2}$, all with $\sigma = 1$ because $1 - 0^n = 1$ for $n \geq 1$. The states that can follow a negative assessment are $W_1, \ldots, W_{k_1}$ and B, all with $\sigma = 0$ because $0^m = 0$ for $m \geq 1$. Initially every record is in G. □

The two implementations therefore generate identical image processes, not only identical stationary statistics. We use $(0, 0)$ in the depth scans and in the stationary analyses, and the one-step state space with $\alpha = \beta = 0$ in the instantaneous panels of Figs. 2a–d and 3.

Table S4. Reference parameters and inertia settings. Values apply unless a figure axis varies them.

Quantity	Value	Varied in
Benefit, cost	$b = 5$, $c = 1$	b/c: Extended Data Figs. 5 and 9
Errors	$e_1 = e_2 = 0.02$	$e_1, e_2 \in [0, 0.45]$: Extended Data Fig. 7
No inertia	$(k_1, k_2) = (0, 0)$; equivalently $\alpha = \beta = 0$	–
Full one-step inertia	$(k_1, k_2) = (1, 1)$, $\alpha = \beta = 1$	(α, β): Extended Data Figs. 3 and 6
Selected multi-step inertia	SC $(2, 0)$, SH $(3, 0)$, SJ $(4, 3)$, SS $(1, 2)$; $\alpha = \beta = 1$	α: Fig. 5, Extended Data Fig. 9
Depth scans	$\alpha = \beta = 1$	k_1, k_2: Figs. 2 and 4; Extended Data Figs. 1, 3 and 8
Initial records	all latent records in G ($r_I = 1$)	all-bad: Extended Data Fig. 2e–h

1.3 The stationary reputation map

Stationary distribution.

Proposition S2. *For $0 < u < 1$ the chain $T(u)$ is irreducible and aperiodic, and its unique stationary distribution is*

$$\pi_{\mathrm{G}} = \frac{u^{k_2+1}}{Z}, \qquad \pi_{\mathrm{W}_m} = v^m \pi_{\mathrm{G}}, \qquad \pi_{\mathrm{B}} = \frac{v^{k_1+1}}{Z}, \qquad \pi_{\mathrm{S}_n} = u^n \pi_{\mathrm{B}}, \qquad Z(u) = v^{k_1} + u^{k_2} - v^{k_1} u^{k_2}, \tag{S5}$$

for $1 \le m \le k_1$ and $1 \le n \le k_2$.

Proof. Irreducibility. From any state, a run of negative assessments reaches B and a run of positive assessments then reaches G. From G every warning state, and from B every recovery state, is reachable. *Aperiodicity.* G has a self-loop with probability $u > 0$. Hence a unique stationary distribution exists.

Paths. The state W_m ($m \ge 1$) is entered only from W_{m-1}, with $\mathrm{W}_0 \equiv \mathrm{G}$, after a negative assessment. Stationarity of W_m requires $\pi_{\mathrm{W}_m} = v\,\pi_{\mathrm{W}_{m-1}}$, hence $\pi_{\mathrm{W}_m} = v^m \pi_{\mathrm{G}}$. Likewise, S_n is entered only from S_{n-1}, with $\mathrm{S}_0 \equiv \mathrm{B}$, after a positive assessment, so $\pi_{\mathrm{S}_n} = u^n \pi_{\mathrm{B}}$.

Flux balance. Partition $\mathscr{L}$ into the good-standing block $\{\mathrm{G}, \mathrm{W}_1, \ldots, \mathrm{W}_{k_1}\}$ and the bad-standing block $\{\mathrm{B}, \mathrm{S}_1, \ldots, \mathrm{S}_{k_2}\}$. The only transition from the first block to the second is the negative assessment from W_{k_1} (from G if $k_1 = 0$), with stationary flux $v\,\pi_{\mathrm{W}_{k_1}} = v^{k_1+1}\pi_{\mathrm{G}}$. The only transition in the opposite direction is the positive assessment from S_{k_2} (from B if $k_2 = 0$), with flux $u^{k_2+1}\pi_{\mathrm{B}}$. In a stationary state the two fluxes are equal, $v^{k_1+1}\pi_{\mathrm{G}} = u^{k_2+1}\pi_{\mathrm{B}}$.

Normalization. Writing $\pi_{\mathrm{G}} = u^{k_2+1}/Z$ and $\pi_{\mathrm{B}} = v^{k_1+1}/Z$ satisfies the flux balance. Summing all probabilities,

$$Z = u^{k_2+1}\sum_{m=0}^{k_1} v^m + v^{k_1+1}\sum_{n=0}^{k_2} u^n = u^{k_2}\left(1 - v^{k_1+1}\right) + v^{k_1}\left(1 - u^{k_2+1}\right) = v^{k_1} + u^{k_2} - v^{k_1}u^{k_2}(u+v),$$

which is the stated Z because $u + v = 1$. □

The exponents record the evidence requirements. Completing a reversal of good standing requires $k_1 + 1$ negative assessments in a row, which suppresses π_{B} by v^{k_1+1}, and completing a recovery requires $k_2 + 1$ positive assessments in a row, which suppresses π_{G} by u^{k_2+1}.

The reputation map. Averaging the image probabilities over Eq. (S5) gives the reputation map of Eq. (10),

$$\mathscr{R}(u) = \pi_{\mathrm{G}} + \sum_{m=1}^{k_1} \alpha^m \pi_{\mathrm{W}_m} + \sum_{n=1}^{k_2} (1-\beta^n)\pi_{\mathrm{S}_n} = \frac{u^{k_2+1}\Sigma_{\mathrm{W}}(v) + v^{k_1+1}\Sigma_{\mathrm{S}}(u)}{Z(u)}, \qquad \Sigma_{\mathrm{W}}(v) = \sum_{m=0}^{k_1}(\alpha v)^m, \quad \Sigma_{\mathrm{S}}(u) = \sum_{n=1}^{k_2} u^n(1-\beta^n). \tag{S6}$$

Empty sums vanish. At the endpoints, the chain is absorbed in B ($u = 0$) or G ($u = 1$), so $\mathscr{R}(0) = 0$ and $\mathscr{R}(1) = 1$, and Eq. (S6) attains these values by continuity. $\mathscr{R}(u)$ is exact for a record with i.i.d. input. How u is determined in a population is the subject of Supplementary Note 1.4. Four special cases are used repeatedly.

1. *No inertia.* For $(k_1, k_2) = (0, 0)$, $Z = 1$ and $\mathscr{R}(u) = u$. By Lemma S1, the same holds for $\alpha = \beta = 0$ at any depths.

2. *One-step inertia*, $(k_1, k_2) = (1, 1)$:

$$\mathscr{R}(u) = \frac{u^2[1 + \alpha(1-u)] + (1-\beta)\,u(1-u)^2}{1 - u + u^2}, \qquad \mathscr{R}(u) - u = \frac{u(1-u)}{1-u+u^2}\left[\alpha u - \beta(1-u)\right]. \tag{S7}$$

   Tolerance raises and strictness lowers the stationary reputation relative to instantaneous updating.

3. *Full strengths*, $\alpha = \beta = 1$: $\Sigma_{\mathrm{W}} = (1 - v^{k_1+1})/u$ and $\Sigma_{\mathrm{S}} = 0$, so

$$\mathscr{R}(u) = \frac{u^{k_2}\left(1 - v^{k_1+1}\right)}{Z(u)}, \qquad 1 - \mathscr{R}(u) = \frac{v^{k_1}\left(1 - u^{k_2+1}\right)}{Z(u)}. \tag{S8}$$

   Exchanging $u \leftrightarrow v$ and $k_1 \leftrightarrow k_2$ exchanges $\mathscr{R} \leftrightarrow 1 - \mathscr{R}$. Tolerance and strictness depths therefore act symmetrically on the two kinds of reputational error.

4. *No recovery path*, $k_2 = 0$: $Z = 1$ and $\mathscr{R}(u) = u\sum_{m=0}^{k_1}(\alpha v)^m$, and β has no effect.

Interpretation through passage times. Let $\Theta_{G\to B}$ be the number of assessments a record starting in G receives until it first enters B, and $\Theta_{B\to G}$ the number from B until it first enters G. Leaving good standing requires a run of k_1+1 negative assessments, and leaving bad standing a run of k_2+1 positive ones. The classical waiting time for a run gives

$$\mathbb{E}[\Theta_{G\to B}] = \frac{1-v^{k_1+1}}{u\,v^{k_1+1}}, \qquad \mathbb{E}[\Theta_{B\to G}] = \frac{1-u^{k_2+1}}{v\,u^{k_2+1}}. \tag{S9}$$

The good-standing block is entered only at G and the bad-standing block only at B, so the record alternates between sojourns of these two kinds. Renewal–reward then gives the stationary probability of good standing, $\mathbb{E}[\Theta_{G\to B}]/(\mathbb{E}[\Theta_{G\to B}]+\mathbb{E}[\Theta_{B\to G}])$, which coincides with Eq. (S8). With full strengths, therefore, $\mathscr{R}(u)$ is the fraction of time spent in good standing, and the depths enter only through the two mean sojourn times. With partial strengths ($\alpha<1$ or $\beta<1$), part of each sojourn is interpreted as the opposite image.

Properties used in the analysis.

Proposition S3. *Let* $0<u<1$.

(a) Monotonicity in the strengths:

$$\partial_\alpha \mathscr{R} = \frac{u^{k_2+1}}{Z}\sum_{m=1}^{k_1} m\,\alpha^{m-1}v^m \geq 0, \qquad \partial_\beta \mathscr{R} = -\frac{v^{k_1+1}}{Z}\sum_{n=1}^{k_2} n\,\beta^{n-1}u^n \leq 0. \tag{S10}$$

(b) Derivative. With $\Omega = u^{k_2+1}\Sigma_W + v^{k_1+1}\Sigma_S$ *the numerator of Eq. (S6),*

$$\begin{aligned}
\mathscr{R}' &= \frac{\Omega' Z - \Omega Z'}{Z^2},\\
\Omega' &= (k_2+1)u^{k_2}\Sigma_W - u^{k_2+1}\sum_{m=1}^{k_1} m\,\alpha^m v^{m-1} - (k_1+1)v^{k_1}\Sigma_S + v^{k_1+1}\sum_{n=1}^{k_2} n\,u^{n-1}(1-\beta^n),\\
Z' &= -k_1 v^{k_1-1}\left(1-u^{k_2}\right) + k_2 u^{k_2-1}\left(1-v^{k_1}\right).
\end{aligned} \tag{S11}$$

(c) Response of a mostly good record ($v\to 0$). If $k_1\geq 1$,

$$1-\mathscr{R}(1-v) = (1-\alpha)\,v + O(v^2), \qquad \mathscr{R}'(1) = 1-\alpha. \tag{S12}$$

If $\alpha = 1$, *for any* $k_1\geq 0$,

$$1-\mathscr{R}(1-v) = \left(1+\sum_{n=1}^{k_2}\beta^n\right)v^{k_1+1} + O(v^{k_1+2}), \qquad \mathscr{R}'(1-v) = (k_1+1)\left(1+\sum_{n=1}^{k_2}\beta^n\right)v^{k_1} + O(v^{k_1+1}). \tag{S13}$$

(d) Response of a mostly bad record ($u\to 0$). If $k_2\geq 1$, $\mathscr{R}(u) = (1-\beta)u + O(u^2)$. *If* $\beta = 1$, *for any* $k_2\geq 0$, $\mathscr{R}(u) = \left(1+\sum_{m=1}^{k_1}\alpha^m\right)u^{k_2+1} + O(u^{k_2+2})$.

Proof. (a) and (b) follow by differentiating Eq. (S6), since Z does not depend on α or β. (c) Write $1-\mathscr{R} = \pi_B + \sum_m(1-\alpha^m)\pi_{W_m} + \sum_n \beta^n \pi_{S_n}$. As $v\to 0$ with $k_1\geq 1$, $Z\to 1$, $\pi_{W_1} = v\pi_G = v+O(v^2)$, and $\pi_B, \pi_{W_{m\geq 2}}, \pi_{S_n} = O(v^2)$, which gives the first statement. If $\alpha = 1$ the warning terms vanish, and $1-\mathscr{R} = \pi_B\left(1+\sum_n \beta^n u^n\right) = v^{k_1+1}\left(1+\sum_n \beta^n\right) + O(v^{k_1+2})$. Differentiating gives the derivative. (d) follows in the same way from $\mathscr{R} = \pi_G\Sigma_W + \sum_n(1-\beta^n)\pi_{S_n}$. □

Parts (c) and (d) quantify the two errors that inertia filters. A good individual that is occasionally assessed negatively (rate v) is interpreted as bad with probability $(1-\alpha)v$ as soon as a warning path exists, irrespective of its length. Only with $\alpha = 1$ does this error become of order v^{k_1+1}. Symmetrically, a bad individual that is occasionally assessed positively is interpreted as good with probability $(1-\beta)u$, or of order u^{k_2+1} when $\beta = 1$. The same expansion implies that, near an all-good state, the marginal reputational penalty for an additional negative assessment is $\mathscr{R}'(1) = 1-\alpha$, which vanishes under full tolerance strength (Supplementary Note 2.3).

Proposition S4 (Concavity without a recovery path)**.** *If* $k_2 = 0$, $\mathscr{R}$ *is concave on* $[0,1]$ *for every* $\alpha\in[0,1]$ *and* $k_1\geq 0$.

Proof. For $k_2 = 0$, $\mathscr{R}(u) = u\sum_{m=0}^{k_1} \alpha^m v^m$. Using $u\sum_{i=0}^{j} v^i = 1 - v^{j+1}$,

$$\mathscr{R}(u) = (1-\alpha)\sum_{j=0}^{k_1-1} \alpha^j \left[1 - v^{j+1}\right] + \alpha^{k_1}\left[1 - v^{k_1+1}\right],$$

because the coefficient of uv^i on the right is $(1-\alpha)\sum_{j=i}^{k_1-1}\alpha^j + \alpha^{k_1} = \alpha^i$. The weights are non-negative, and each $1-(1-u)^{j+1}$ is concave in u. □

Proposition S4 covers the selected SC $(2,0)$ and SH $(3,0)$ architectures. With a recovery path and $\beta = 1$, $\mathscr{R}$ is not concave. It starts as u^{k_2+1} near $u = 0$, and for $k_1 \geq 1$ and $\alpha = 1$ it approaches 1 as $1 - c\,v^{k_1+1}$ near $u = 1$.

1.4 Private reputations in the population

Reputation quantities and mean-field closure. At fixed strategy composition $\boldsymbol{f}$, let r_I be the probability that a randomly chosen observer regards a randomly chosen type-I individual as good, that is, the expected image probability $\sigma(L_{ij})$ over observers i and type-I targets j, and let $\bar{r} = \sum_I f_I r_I$. Because all individuals use the same norm and record mechanism, all observers are statistically equivalent in the mean-field description. Two assumptions turn the single-record result into population equations. Both are standard in analyses of private assessment[1–5].

(A1) *Conditional independence of images.* The images that the donor and the observer hold of a recipient are independent given the recipient's strategy.

(A2) *Independent inputs.* The successive assessments entering any record about a type-I target are independent, each positive with the same probability u_I.

(A1) neglects correlations between the records that different observers hold of the same individual. Such correlations arise because two observers may have assessed the same past actions of that individual, and because targets of the same strategy differ in their histories. (A2) neglects temporal correlations in the inputs to a record. The finite-population simulations retain both effects and are used to test reputation formation without the closure (Supplementary Note 3).

Positive-assessment probabilities and self-consistency. Under (A1), donor and observer both regard a random recipient as good with probability $h_{\mathrm{GG}} = \sum_J f_J r_J^2$, both regard it as bad with probability $h_{\mathrm{BB}} = \sum_J f_J (1-r_J)^2$, and hold a given ordered disagreement with probability $h_{\mathrm{GB}} = \sum_J f_J r_J (1-r_J)$, with $h_{\mathrm{GG}} + 2h_{\mathrm{GB}} + h_{\mathrm{BB}} = 1$. Conditioning on the recipient's strategy matters, because $h_{\mathrm{GG}} \geq \bar{r}^2$, with equality only if all strategies have the same reputation. An ALLC donor intends C whatever the recipient and an ALLD donor intends D, so their assessments depend only on the observer's image. A DISC donor intends C if its own image is G. Supplementary Table S5 lists the four cases, and summing them gives the positive-assessment probabilities u_I of Eq. (11). Using Eq. (S3), the DISC expression can be written as $u_{\mathrm{DISC}} = \sum_J f_J u_{\mathrm{R}}(r_J)$. A DISC donor is thus assessed as if it were a resident in a homogeneous population whose reputation equals that of its recipient, averaged over recipients. Under (A2), each record about a type-I target has the stationary response $\mathscr{R}(u_I)$, which gives the self-consistency equations (12). Because u_{ALLC} and u_{ALLD} depend on $\boldsymbol{r}$ only through $\bar{r}$, these equations form a three-dimensional fixed-point problem that can have several solutions.

Table S5. Assessment of a DISC donor. The first image is the donor's, the second the observer's.

Images (donor, observer)	Probability	Intended action	Positive assessment
(G, G)	h_{GG}	C	$P^+_{\mathrm{C\|G}}$
(G, B)	h_{GB}	C	$P^+_{\mathrm{C\|B}}$
(B, G)	h_{GB}	D	$P^+_{\mathrm{D\|G}}$
(B, B)	h_{BB}	D	$P^+_{\mathrm{D\|B}}$

Full latent-state dynamics. Within the same closure, let $\boldsymbol{\pi}_I(t')$ be the row vector of latent-state probabilities of records about type-I targets, with time t' measured in assessments per record. Asynchronous updating, in which each record is updated at unit rate, gives

$$\frac{\mathrm{d}\boldsymbol{\pi}_I}{\mathrm{d}t'} = \boldsymbol{\pi}_I\left[T\left(u_I(\boldsymbol{r};\boldsymbol{f})\right) - \mathbf{I}\right], \qquad r_I = \boldsymbol{\pi}_I\boldsymbol{\sigma}. \tag{S14}$$

This flow conserves probability and non-negativity. It is the mean-field limit of the finite simulation when time is measured in sweeps and assessment probabilities are replaced by their closed forms.

Proposition S5 (Stationary equivalence). *Suppose* $0 < u_I < 1$ *at a stationary state. Then* $\{\boldsymbol{\pi}_I^*\}$ *is a stationary state of Eq.* (S14) *if and only if* $\boldsymbol{\pi}_I^* = \boldsymbol{\pi}(u_I^*)$ *with* $u_I^* = u_I(\boldsymbol{r}^*; \boldsymbol{f})$ *and* $\boldsymbol{r}^*$ *solves Eq.* (12).

Proof. At a stationary state, $\boldsymbol{\pi}_I^* T(u_I^*) = \boldsymbol{\pi}_I^*$, so $\boldsymbol{\pi}_I^*$ is the unique stationary distribution of $T(u_I^*)$ (Proposition S2), and $r_I^* = \boldsymbol{\pi}_I^* \boldsymbol{\sigma} = \mathscr{R}(u_I^*)$. Conversely, given a solution $\boldsymbol{r}^*$ of Eq. (12), the distributions $\boldsymbol{\pi}(u_I^*)$ are stationary and reproduce $\boldsymbol{r}^*$. □

Transient dynamics are not reduced in the same way. Projecting Eq. (S14) gives $\mathrm{d}r_I/\mathrm{d}t' = \boldsymbol{\pi}_I T(u_I)\boldsymbol{\sigma} - r_I$, and $\boldsymbol{\pi}_I T(u_I)\boldsymbol{\sigma}$ (the expected image probability after one more update) generally differs from $\mathscr{R}(u_I)$. For example, for one-step inertia with $\alpha = \beta = 1$ and all records in G, $\boldsymbol{\pi}_I T(u)\boldsymbol{\sigma} = 1$, whereas $\mathscr{R}(u) = u^2(2-u)/(1-u+u^2) < 1$. Two populations with the same r_I but different amounts of accumulated evidence evolve differently.

Branch selection. To select a stationary solution we use the auxiliary flow (13), whose fixed points are exactly the solutions of Eq. (12) and whose Jacobian is $\mathrm{diag}[\mathscr{R}'(u_I)]\,\partial \boldsymbol{u}/\partial \boldsymbol{r} - \mathbf{I}$. The time τ is a computational device. Equation (13) is not the transient dynamics of the records, and its stability need not coincide with that of Eq. (S14) in general. Three selection protocols are used.

1. *Main convention* (all strategy-dynamics results). At every composition, Eq. (13) is integrated from $r_I(0) = 1$, which corresponds to all records in G. The replicator field is thus defined on the branch reached from good standing.

2. *Highest strictly stable root* (homogeneous DISC in Extended Data Figs. 2 and 3, local invasion and Fig. 5). This is the highest root of $F(r)$ with $F'(r) < 0$. It coincides with protocol 1 unless the highest root is a fold, which is marginal.

3. *Quasi-static tracking* (Extended Data Fig. 2e–h). Reputations are initialized once, at $t = 0$, as all-good or all-bad and then follow the strategy trajectory, Eq. (13) being re-solved from the current reputations at every evaluation. A branch is then left only where it ends at a fold, so the outcome can depend on reputational history.

Homogeneous discriminator populations. In a homogeneous DISC population, $h_{\mathrm{GG}} = r^2$, $h_{\mathrm{GB}} = r(1-r)$ and $h_{\mathrm{BB}} = (1-r)^2$, and u_{DISC} reduces to $u_{\mathrm{R}}(r)$ of Eq. (S3). The stationary resident reputations are the roots of

$$F(r) = \mathscr{R}\big[u_{\mathrm{R}}(r)\big] - r, \qquad F'(r) = \mathscr{R}'\big[u_{\mathrm{R}}(r)\big]\, u_{\mathrm{R}}'(r) - 1. \tag{S15}$$

A simple root is stable in the scalar flow $\mathrm{d}r/\mathrm{d}\tau = F(r)$ if $F' < 0$ and unstable if $F' > 0$. Because $F(0) \geq 0$ and $F(1) \leq 0$, roots alternate in stability, and a pair of roots appears or disappears at a fold, $F = F' = 0$. Bistability (Extended Data Figs. 2 and 3) means two stable roots separated by an unstable one. With errors, $F(1) < 0$, so the highest root is approached from $r = 1$. For every root in the depth planes with $\alpha = \beta = 1$ ($0 \leq k_1, k_2 \leq 8$), the classification by F' agreed with the leading eigenvalue of Eq. (S14) linearized about $\boldsymbol{\pi}(u_{\mathrm{R}}(r))$ on the probability simplex. This linearization includes the dependence of u_{R} on the current image probabilities. On a regular stable branch ($F' < 0$), implicit differentiation of $F(r^*; \theta) = 0$ for $\theta \in \{\alpha, \beta\}$ gives

$$\frac{\partial r^*}{\partial \theta} = -\frac{\partial_\theta \mathscr{R}\big[u_{\mathrm{R}}(r^*)\big]}{F'(r^*)}, \tag{S16}$$

so by Eq. (S10) the resident reputation increases with α and decreases with β along a stable branch. The derivative diverges at a fold, where the branch ends.

The ALLD–DISC edge. On the edge $f_{\mathrm{ALLC}} = 0$, $f_{\mathrm{DISC}} = x$, $f_{\mathrm{ALLD}} = 1 - x$, the three-dimensional problem reduces exactly to one equation. Given $\bar{r}$, $r_{\mathrm{ALLD}} = \mathscr{R}[\Phi_{\mathrm{D}}(\bar{r})]$ and $r_{\mathrm{ALLC}} = \mathscr{R}[\Phi_{\mathrm{C}}(\bar{r})]$ follow directly, and $r_{\mathrm{DISC}} = [\bar{r} - (1-x) r_{\mathrm{ALLD}}]/x$ follows from the definition of $\bar{r}$. Substituting these into $r_{\mathrm{DISC}} = \mathscr{R}[x\, u_{\mathrm{R}}(r_{\mathrm{DISC}}) + (1-x)\, u_{\mathrm{R}}(r_{\mathrm{ALLD}})]$ gives one equation in $\bar{r}$. Its roots with $r_{\mathrm{DISC}} \in [0, 1]$ are all self-consistent states on the edge, and their stability is that of Eq. (13). The normalized payoff advantage of DISC over ALLD on the edge is $(\Pi_{\mathrm{DISC}} - \Pi_{\mathrm{ALLD}})/[c(1-e_1)] = (b/c)\, x\, (r^*_{\mathrm{DISC}} - r^*_{\mathrm{ALLD}}) - \bar{r}^*$. Where it changes sign continuously, the edge has an equilibrium. Where the selected branch ends at a fold, the sign can change discontinuously (Extended Data Fig. 4).

1.5 Strategy dynamics and cooperation

Payoffs and replicator dynamics. Reputations are assumed to reach their stationary state quickly compared with changes in strategy frequencies, as in previous models of indirect reciprocity[2–4,6]. Payoffs are computed per unit time, in which each individual acts on average once as donor and once as recipient. A type-I recipient receives intended help from ALLC donors (probability f_{ALLC}) and from DISC donors that regard it as good (probability $f_{\mathrm{DISC}} r_I^*$). As a donor, ALLC intends to help with probability 1, ALLD with probability 0, and DISC with probability $\bar{r}^*$, because its recipient is a random individual. Each intended donation is realized with probability $1 - e_1$, so

$$\Pi_I = (1-e_1)\big[b\,(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}}\, r_I^*) - c\,\chi_I\big], \qquad \chi_{\mathrm{ALLC}} = 1, \quad \chi_{\mathrm{ALLD}} = 0, \quad \chi_{\mathrm{DISC}} = \bar{r}^*, \tag{S17}$$

which gives Eq. (15). Every realized donation creates one benefit and one cost, so $\bar{\Pi} = \sum_I f_I \Pi_I = (b-c)\rho$. The factor $1-e_1$ rescales time in the replicator dynamics (16), but e_1 also changes the reputations through $P^{+}_{\mathrm{A}|\mathrm{Y}}$. Edges and vertices of the simplex are invariant. At a vertex where all individuals use J, the eigenvalues are the growth rates $\Pi_I - \Pi_J$ of the two absent strategies, evaluated with the reputations of that vertex. At the DISC vertex these are the invasion payoffs of Supplementary Note 2. At an equilibrium on an edge, the tangent eigenvalue describes stability within the edge and the transverse eigenvalue is the growth rate of the absent strategy. Interior equilibria are classified by the Jacobian of the replicator dynamics on the simplex. Because reputations can jump where the selected branch ends, the replicator field can be discontinuous at folds (Fig. 3 and Extended Data Fig. 4).

Proposition S6 (Line of equilibria for Scoring without inertia)**.** *Under Scoring with* $\mathscr{R}(u) = u$, $\Pi_{\mathrm{ALLC}} = \Pi_{\mathrm{ALLD}} = \Pi_{\mathrm{DISC}}$ *at every composition with* $f_{\mathrm{DISC}} = f^{\dagger} \equiv c/[b(\varepsilon - e_2)]$.

Proof. Under SC, $\Phi_{\mathrm{C}} \equiv \varepsilon$ and $\Phi_{\mathrm{D}} \equiv e_2$ do not depend on the recipient, so $u_{\mathrm{DISC}} = \varepsilon(h_{\mathrm{GG}} + h_{\mathrm{GB}}) + e_2(h_{\mathrm{GB}} + h_{\mathrm{BB}}) = e_2 + (\varepsilon - e_2)\bar{r}$, and $r_{\mathrm{ALLC}} = \varepsilon$, $r_{\mathrm{ALLD}} = e_2$, $r_{\mathrm{DISC}} = e_2 + (\varepsilon - e_2)\bar{r}$. Then $\Pi_{\mathrm{ALLC}} - \Pi_{\mathrm{DISC}} = (1-e_1)(1-\bar{r})[b f_{\mathrm{DISC}}(\varepsilon - e_2) - c]$ and $\Pi_{\mathrm{ALLD}} - \Pi_{\mathrm{DISC}} = (1-e_1)\bar{r}[c - b f_{\mathrm{DISC}}(\varepsilon - e_2)]$, which both vanish at $f_{\mathrm{DISC}} = f^{\dagger}$. □

At the reference parameters, $f^{\dagger} = 1/(5 \times 0.9408) \approx 0.2126$. Along this line every composition is an equilibrium, so the line is neutral in its own direction. Its stability in the transverse direction follows from the same payoff differences. Off the line, $\mathrm{d}f_{\mathrm{DISC}}/\mathrm{d}t = (1-e_1) f_{\mathrm{DISC}} [b f_{\mathrm{DISC}}(\varepsilon - e_2) - c] [f_{\mathrm{ALLD}}\bar{r} - f_{\mathrm{ALLC}}(1-\bar{r})]$. The line therefore attracts nearby trajectories where $f_{\mathrm{ALLC}}(1-\bar{r}) > f_{\mathrm{ALLD}}\bar{r}$, that is, on the part closer to the ALLC–DISC edge, and repels them on the part closer to the ALLD–DISC edge. At the reference parameters the two parts meet at $f_{\mathrm{ALLC}} \approx 0.266$, where intended cooperation equals $\bar{r} \approx 0.34$. On the attracting part, intended cooperation increases to 0.99 at the ALLC–DISC edge. Trajectories that reach the line drift neutrally along it, which is why integration under SC without inertia sometimes reaches the time cap without affecting the outcome class.

Cooperation measures. At a composition, intended and realized cooperation are ρ_{int} and ρ, and in a homogeneous DISC population $\rho_{\mathrm{int}} = r^{*}_{\mathrm{R}}$. Across all scans, every final state belonged to one of two classes. The first is the zero-cooperation set, comprising the ALLD vertex together with the low-reputation segment of the ALLD–DISC edge, on which DISC regard one another as bad, earn slightly less than ALLD and drift slowly towards ALLD (final $\rho \leq 7 \times 10^{-4}$). The second is an attractor with positive cooperation (final $\rho \geq 0.34$, and between 0.974 and 0.980 in nearly all conditions). The cooperative basin is the set of initial compositions whose trajectories reach the second class. With $\bar{\rho}_{\mathrm{D}}$ the mean final cooperation of the remaining trajectories,

$$\bar{\rho} = \omega_{\mathrm{C}}\bar{\rho}_{\mathrm{C}} + (1-\omega_{\mathrm{C}})\bar{\rho}_{\mathrm{D}} \approx \omega_{\mathrm{C}}\bar{\rho}_{\mathrm{C}}, \tag{S18}$$

so a change in $\bar{\rho}$ can be attributed to the basin size and to the cooperation level within it. At the reference parameters, $\bar{\rho}_{\mathrm{C}}$ is 0.975–0.980 for one-step and selected multi-step inertia under all norms, limited by implementation error. At least 94% of the increase in $\bar{\rho}$ produced by the selected architectures comes from the enlarged basin. Supplementary Table S6 lists the basin sizes and mean cooperation. These quantities depend on the area-uniform sampling of initial compositions and are not probabilities over any other distribution of starting points.

Table S6. Cooperative basins and mean cooperation at the reference parameters. Basin size ω_{C} and mean realized cooperation $\bar{\rho}$ over 512 area-uniform initial compositions, for instantaneous updating, full one-step inertia and the selected multi-step architectures ($b/c = 5$, $e_1 = e_2 = 0.02$, $\alpha = \beta = 1$).

	No inertia		One-step $(1,1)$		Selected multi-step	
Norm	ω_{C}	$\bar{\rho}$	ω_{C}	$\bar{\rho}$	ω_{C}	$\bar{\rho}$
SC	0.635	0.610	0.605	0.591	0.883	0.865
SH	0	0	0.078	0.076	0.705	0.691
SJ	0	0	0.230	0.225	0.713	0.699
SS	0.561	0.543	0.713	0.696	0.760	0.741

Supplementary Note 2. Invasion of a discriminator population

2.1 Rare mutants

Consider a homogeneous DISC population whose reputation is the selected root r^{*}_{R}, and a rare mutant X that uses the residents' norm and record mechanism. Because X is rare, its recipients, its observers and the individuals it is assessed against are

residents. Residents regard each other as good with probability r_R^*, and their assessment environment is unchanged. We write $\Phi_D^* = r_R^* P_{D|G}^+ + (1-r_R^*)P_{D|B}^+$ and $\Phi_C^* = r_R^* P_{C|G}^+ + (1-r_R^*)P_{C|B}^+$.

Lemma S7 (Rare-mutant identity)**.** *Suppose that X intends to help a resident recipient with probability χ_X, independently of the observer's image of that recipient and independently across interactions. Then X is assessed positively with probability*

$$u_X = \chi_X \Phi_C^* + (1-\chi_X)\Phi_D^* = u_R^* - (r_R^* - \chi_X)\,\Delta\Phi^*, \tag{S19}$$

its stationary reputation is $r_X^* = \mathscr{R}(u_X)$, and its normalized payoff advantage Λ_X is given by Eq. (17).

Proof. The observer regards the resident recipient as good with probability r_R^*, independently of X's intention, so X's assessment probability is the mixture of Φ_C^* and Φ_D^* with weights χ_X and $1-\chi_X$. Independent actions imply independent assessments, so (A2) holds exactly for X's records in the fixed resident environment, and $r_X^* = \mathscr{R}(u_X)$. A resident receives help from residents with probability r_R^* and helps with probability r_R^*, so $\Pi_R = (1-e_1)(b-c)r_R^*$. The mutant receives help with probability r_X^* and helps with probability χ_X, so $\Pi_X = (1-e_1)(br_X^* - c\chi_X)$. Subtracting and dividing by $c(1-e_1) > 0$ gives Eq. (17). The second form of Eq. (S19) uses $u_R^* = r_R^*\Phi_C^* + (1-r_R^*)\Phi_D^*$. □

The resident corresponds to $\chi = r_R^*$, ALLD to $\chi = 0$, ALLC to $\chi = 1$ and the opportunistic discriminator to $\chi = (1-\delta)r_R^*$. All of these mutants therefore lie on the single curve $\chi \mapsto \mathscr{R}[u_R^* - (r_R^* - \chi)\Delta\Phi^*]$, which is $\mathscr{R}$ evaluated along the segment between Φ_D^* ($\chi = 0$) and Φ_C^* ($\chi = 1$), passing through u_R^* at $\chi = r_R^*$.

2.2 Unconditional mutants

The reputations of rare ALLD and ALLC are $r_{ALLD}^* = \mathscr{R}(\Phi_D^*)$ and $r_{ALLC}^* = \mathscr{R}(\Phi_C^*)$. These are reputations of rare mutants in the resident population, not of homogeneous ALLC or ALLD populations. Lemma S7 gives

$$\lambda_{ALLD} = (1-e_1)\big[c\,r_R^* - b\,(r_R^* - r_{ALLD}^*)\big], \qquad \lambda_{ALLC} = (1-e_1)\big[b\,(r_{ALLC}^* - r_R^*) - c\,(1-r_R^*)\big], \tag{S20}$$

where $\lambda_X = \Pi_X - \Pi_R$. ALLD saves the resident's donation cost cr_R^* but may be helped less. ALLC can be regarded better than residents because it also helps recipients that residents regard as bad, but it pays for these donations.

Proposition S8 (Resistance to ALLC and ALLD)**.**

(a) If $r_R^* > r_{ALLD}^*$, rare ALLD is excluded ($\lambda_{ALLD} < 0$) if and only if $b/c > R_L = r_R^*/(r_R^* - r_{ALLD}^*)$. If $r_R^* \le r_{ALLD}^*$ and $r_R^* > 0$, no b/c excludes ALLD.

(b) If $r_{ALLC}^* > r_R^*$, rare ALLC is excluded ($\lambda_{ALLC} < 0$) if and only if $b/c < R_U = (1-r_R^*)/(r_{ALLC}^* - r_R^*)$. If $r_{ALLC}^* \le r_R^* < 1$, ALLC is excluded for every b/c.

(c) If $r_R^* > r_{ALLD}^*$ and $r_{ALLC}^* > r_R^*$, the residents resist both mutants if and only if $R_L < b/c < R_U$, which is Eq. (1) of the main text.

Proof. Rearrange the signs of Eq. (S20). $\lambda_{ALLD} < 0$ if and only if $b(r_R^* - r_{ALLD}^*) > cr_R^*$, and $\lambda_{ALLC} < 0$ if and only if $b(r_{ALLC}^* - r_R^*) < c(1-r_R^*)$. Divide by the positive quantities stated in each case. □

When is the window open?

Proposition S9 (Chord condition)**.** *Under the conditions of Proposition S8(c), the window is non-empty if and only if*

$$r_R^*\,\mathscr{R}(\Phi_C^*) + (1-r_R^*)\,\mathscr{R}(\Phi_D^*) < \mathscr{R}(u_R^*). \tag{S21}$$

Proof. $R_L < R_U$ is equivalent to $r_R^*(r_{ALLC}^* - r_R^*) < (1-r_R^*)(r_R^* - r_{ALLD}^*)$, that is, $r_R^* r_{ALLC}^* + (1-r_R^*)r_{ALLD}^* < r_R^*$. Self-consistency gives $r_R^* = \mathscr{R}(u_R^*)$. □

The resident's own action is a mixture of the actions of ALLC and ALLD with weights r_R^* and $1-r_R^*$, and its assessment probability u_R^* is the corresponding mixture of Φ_C^* and Φ_D^*. The window is open precisely when the reputation of this mixture exceeds the same mixture of the two pure reputations, that is, when $\mathscr{R}$ lies above its chord between Φ_D^* and Φ_C^* at u_R^*.

Corollary S10 (No window without inertia)**.** *If $\mathscr{R}(u) = u$ and $\Delta\Phi^* > 0$, then $R_L = R_U = 1/\Delta\Phi^*$. Resistance to ALLD requires $b/c > 1/\Delta\Phi^*$ and resistance to ALLC requires $b/c < 1/\Delta\Phi^*$, so no benefit-to-cost ratio gives resistance to both.*

Proof. With $\mathscr{R}(u) = u$, $r_R^* = u_R^*$, $r_R^* - r_{ALLD}^* = r_R^*\Delta\Phi^*$ and $r_{ALLC}^* - r_R^* = (1-r_R^*)\Delta\Phi^*$. □

If $\Delta\Phi^* \le 0$ without inertia (SJ, where the instantaneous resident has $r_R^* = 1/2$ and $\Delta\Phi^* = 0$), ALLD is regarded at least as well as the residents and cannot be excluded. The thresholds also have a marginal interpretation. Define the reputation lost per unit of withheld cooperation, $H_{\mathrm{ALLD}} = (r_R^* - r_{\mathrm{ALLD}}^*)/r_R^*$, and the reputation gained per unit of extra cooperation, $H_{\mathrm{ALLC}} = (r_{\mathrm{ALLC}}^* - r_R^*)/(1 - r_R^*)$. Then $R_L = 1/H_{\mathrm{ALLD}}$ and $R_U = 1/H_{\mathrm{ALLC}}$, and the window reads $H_{\mathrm{ALLC}} < c/b < H_{\mathrm{ALLD}}$. Cooperating more than the residents must not pay off, and cooperating less must not pay off either. Because $\Phi_C^* - u_R^* = (1 - r_R^*)\Delta\Phi^*$ and $u_R^* - \Phi_D^* = r_R^*\Delta\Phi^*$, H_{ALLC} and H_{ALLD} are $\Delta\Phi^*$ times the slopes of the chords of $\mathscr{R}$ on $[u_R^*, \Phi_C^*]$ and $[\Phi_D^*, u_R^*]$, respectively. Inertia opens the window by making $\mathscr{R}$ flatter above the resident's assessment probability than below it, so that extra cooperation buys little additional reputation while withheld cooperation is still penalized. Supplementary Table S7 lists the values at the reference parameters, and Supplementary Note 4 gives the numerical classification rule.

Table S7. Reference values for a resident discriminator population. Highest stable homogeneous-DISC root r_R^*, reputations of rare ALLD and ALLC, and the thresholds of Eq. (1); $e_1 = e_2 = 0.02$, and $\alpha = \beta = 1$ for inertial settings. Without inertia $R_L = R_U$ (Corollary S10). Under SJ without inertia, ALLD is regarded as well as residents and cannot be excluded.

Norm	Setting	r_R^*	r_{ALLD}^*	r_{ALLC}^*	R_L	R_U
SC	none $(0,0)$	0.337838	0.020000	0.960800	1.0629	1.0629
	one-step $(1,1)$	0.996279	0.000808	0.996869	1.0008	6.3070
	selected $(2,0)$	0.999940	0.058808	0.999940	1.0625	230.24
SH	none $(0,0)$	0.020391	0.020000	0.039184	52.127	52.127
	one-step $(1,1)$	0.995272	0.000808	0.996111	1.0008	5.6409
	selected $(3,0)$	0.999998	0.077632	0.999998	1.0842	4410.4
SJ	none $(0,0)$	0.500000	0.500000	0.500000	–	–
	one-step $(1,1)$	0.995307	0.001215	0.996133	1.0012	5.6865
	selected $(4,3)$	0.9999996	8.3×10^{-7}	0.9999996	1.0000	20910
SS	none $(0,0)$	0.833333	0.180000	0.964000	1.2755	1.2755
	one-step $(1,1)$	0.996297	0.001122	0.996880	1.0011	6.3441
	selected $(1,2)$	0.993660	0.000036	0.995245	1.0000	4.0001

2.3 Opportunistic discriminators and secure tolerance strengths

The mutant family. An opportunistic discriminator (O-DISC) uses the residents' norm and record mechanism and follows DISC, except that whenever its own image of the recipient is G it independently replaces the intended cooperation by intended defection with probability $\delta \in [0,1]$. This choice precedes implementation error, and O-DISC never cooperates with recipients it regards as bad. Its images of resident recipients are good with probability r_R^* and are independent of the observer's images by (A1), so it intends to help with probability $\chi = (1-\delta)r_R^*$. Lemma S7 gives the assessment probability $u_X(\delta)$ of Eq. (18) and the invasion payoff $\Lambda(\delta)$ of Eq. (2). Equivalently, $u_X(\delta) = (1-\delta)u_R^* + \delta\Phi_D^*$, so the mutant's assessment probability moves linearly from the resident's ($\delta = 0$) to that of ALLD ($\delta = 1$). Hence $\Lambda(0) = 0$, the mutant being a neutral copy of a resident, and $\Lambda(1) = \lambda_{\mathrm{ALLD}}/[c(1-e_1)]$. The family contains neither ALLC nor mutants whose omissions are correlated in time.

Weak opportunism. Differentiating Eq. (2) with $\mathrm{d}r_X^*/\mathrm{d}\delta = -r_R^*\Delta\Phi^*\mathscr{R}'[u_X(\delta)]$ gives Eq. (3). If $\Lambda'(0) > 0$, all sufficiently weak opportunists invade. If $\Lambda'(0) < 0$, they are excluded, but larger δ may still invade. The product $\Delta\Phi^*\mathscr{R}'(u_R^*)$ is the reputation lost per unit of withheld cooperation for an infinitesimal deviation. The norm converts withheld cooperation into a lower positive-assessment probability at rate $\Delta\Phi^*$, and the record converts this into a lower reputation at rate $\mathscr{R}'(u_R^*)$.

Resistance to the whole family. For $r_R^* > 0$ and $0 < \delta \le 1$, the reputation lost per unit of withheld cooperation can be written as

$$H(\delta) = \frac{r_R^* - r_X^*(\delta)}{\delta\, r_R^*} = \Delta\Phi^* \frac{\mathscr{R}(u_R^*) - \mathscr{R}[u_X(\delta)]}{u_R^* - u_X(\delta)} = \Delta\Phi^* \int_0^1 \mathscr{R}'\left(u_R^* - s\,\delta\, r_R^*\,\Delta\Phi^*\right) \mathrm{d}s, \tag{S22}$$

with continuous extension $H(0) = \Delta\Phi^*\mathscr{R}'(u_R^*)$. The second form uses $r_R^* = \mathscr{R}(u_R^*)$ and $u_R^* - u_X = \delta r_R^*\Delta\Phi^*$ (for $\Delta\Phi^* \neq 0$). $H(\delta)$ is $\Delta\Phi^*$ times the slope of the chord of $\mathscr{R}$ between $u_X(\delta)$ and u_R^*. Then $\Lambda(\delta) = \delta r_R^*[1 - (b/c)H(\delta)]$, and $H(1) = H_{\mathrm{ALLD}}$.

Proposition S11 (Family criterion)**.** *Let $r_R^* > 0$ and $H_{\min} = \min_{0\le\delta\le1} H(\delta)$. Then $\Lambda(\delta) \le 0$ for every $\delta \in (0,1]$ if and only if $H_{\min} \ge c/b$, which is Eq. (4) of the main text. If $H_{\min} > c/b$, every O-DISC mutant is strictly excluded. If $H_{\min} < c/b$, a mutant with δ in a non-empty interval invades.*

Proof. For $\delta > 0$, $\Lambda(\delta)$ has the sign of $1 - (b/c)H(\delta)$. The minimum includes $\delta = 0$ through the continuous extension, which is required because $\Lambda'(0)$ has the sign of $1 - (b/c)H(0)$. If $H(0) < c/b$, all small $\delta > 0$ invade although no single $\delta > 0$ attains $H(0)$. Continuity of H gives the interval. □

Corollary S12 (Curvature and the location of the worst mutant)**.** *Let* $\Delta\Phi^* > 0$.

(a) If $\mathscr{R}$ *is concave on* $[\Phi_D^*, u_R^*]$*, then* H *is non-decreasing in* δ*,* $H_{\min} = H(0)$*, and the family is excluded if and only if* $\Lambda'(0) \le 0$*. This holds for every architecture with* $k_2 = 0$ *(Proposition S4).*

(b) If $\mathscr{R}$ *is convex on* $[\Phi_D^*, u_R^*]$*, then* $H_{\min} = H(1)$*, and resistance to ALLD implies resistance to the whole family.*

(c) In general, the worst mutant can be interior, and all δ *must be checked.*

Proof. As δ increases, $u_X(\delta)$ decreases from u_R^* to Φ_D^*. For concave $\mathscr{R}$, the slope of the chord from u_X to the fixed right endpoint u_R^* is non-increasing in u_X, hence non-decreasing in δ. For convex $\mathscr{R}$ it is non-increasing in δ. □

Part (a) also shows how tolerance creates vulnerability. With a concave response, the reputation lost by a mutant is at most the linear interpolation between the resident and ALLD, $r_R^* - r_X^*(\delta) \le \delta(r_R^* - r_{ALLD}^*)$, so $\Lambda(\delta) \ge \delta\Lambda(1)$. Excluding ALLD then says little about partial opportunism. Near an all-good resident, Eq. (S12) gives $\mathscr{R}'(u_R^*) \approx 1-\alpha$ for $k_1 \ge 1$. With full tolerance strength, Eq. (S13) makes $\mathscr{R}'(u_R^*)$ of order $(1-u_R^*)^{k_1}$, so $H(0)$ and the marginal penalty vanish as the warning path lengthens. For SJ $(4,3)$ with $\beta = 1$ at the reference parameters, $H(0) = 4.8 \times 10^{-5}$ at $\alpha = 1$ but 0.2037 at $\alpha = 0.8$, where $c/b = 0.2$.

An exact threshold in the error-free limit.

Proposition S13 (Error-free security threshold)**.** *Let* $e_1 = e_2 = 0$*,* $\beta = 1$ *and* $k_1 \ge 1$*, and consider the all-good resident* $r_R^* = 1$ *under any of the four norms and any* $k_2 \ge 0$*. Every O-DISC mutant is excluded,* $\Lambda(\delta) \le 0$ *for all* $\delta \in (0,1]$*, if and only if*

$$\alpha \le 1 - \frac{c}{b}. \tag{S23}$$

Proof. Without errors, $P_{C|G}^+ = 1$ and $P_{D|G}^+ = 0$, so $r = 1$ is a resident root with $u_R^* = 1$ and $\Delta\Phi^* = 1$, and $u_X(\delta) = 1-\delta$. *Necessity.* By Eq. (S12), $\mathscr{R}'(1) = 1-\alpha$, so $\Lambda'(0) = 1-(b/c)(1-\alpha)$, which is positive if $\alpha > 1 - c/b$. *Sufficiency.* With $\beta = 1$ the recovery term of Eq. (S6) vanishes, and for $0 < u \le 1$

$$\frac{\mathscr{R}_{k_1,k_2}(u)}{\mathscr{R}_{k_1,0}(u)} = \frac{u^{k_2}}{u^{k_2} + v^{k_1}\left(1-u^{k_2}\right)} \le 1,$$

where subscripts denote depths. $\mathscr{R}_{k_1,0}$ is concave (Proposition S4) with $\mathscr{R}_{k_1,0}(1) = 1$ and $\mathscr{R}'_{k_1,0}(1) = 1-\alpha$, so it lies below its tangent at $u = 1$: $\mathscr{R}_{k_1,k_2}(u) \le \mathscr{R}_{k_1,0}(u) \le 1-(1-\alpha)(1-u)$. Hence $\Lambda(\delta) = \delta - (b/c)[1-\mathscr{R}_{k_1,k_2}(1-\delta)] \le \delta[1-(b/c)(1-\alpha)] \le 0$ when $\alpha \le 1-c/b$. □

At $b/c = 5$ the threshold is $\alpha = 0.8$, the tolerance strength used for illustration in Fig. 5 and Extended Data Fig. 9. With errors, the resident is no longer exactly all-good, $\Delta\Phi^* < 1$, and the boundary moves (for example to $\alpha_U = 0.8036$ under SJ $(4,3)$). The result is conditional on the all-good resident being selected. Stability of $r = 1$ itself requires $F'(1) = (1-\alpha)(2-p) - 1 < 0$, that is, $\alpha > 0$ for SC and SS and $\alpha > 1/2$ for SH and SJ.

Secure tolerance intervals. Fix the norm, depths, β, errors and b/c. A tolerance strength α is secure if the high-reputation resident exists as a strictly stable root ($F = 0$, $F' < 0$) and satisfies $(b/c)H_{\min} \ge 1$. The secure set can be empty, can be the whole of $[0,1]$, or can consist of intervals. Its boundaries are of three kinds.

1. *Fold (lower boundary* α_L*).* The high-reputation branch ends where $F(r,\alpha) = 0$ and $\partial_r F(r,\alpha) = 0$, with $F(r,\alpha) = \mathscr{R}[u_R(r);\alpha,\beta,k_1,k_2] - r$, equivalently $\mathscr{R}'[u_R(r)]\,u_R'(r) = 1$. The fold is marginally stable and is excluded. Because these equations contain no payoff parameter, α_L does not depend on b/c. By Eq. (S16), the resident reputation increases with α along the stable branch, and the branch is lost when α is lowered through α_L. In all cases computed, the resident then falls to a low-reputation root that opportunists invade.

2. *Invasion (upper boundary* α_U*).* $(b/c)H_{\min} = 1$ on the high branch. In every case we computed with an interior upper boundary, the minimum was attained at $\delta \to 0$, so that

$$F(r,\alpha_U) = 0, \qquad \frac{b}{c}\,\Delta\Phi(r)\,\mathscr{R}'\left[u_R(r);\alpha_U\right] = 1. \tag{S24}$$

   For $k_2 = 0$ this is exact by Corollary S12(a). The boundary is included (neutral mutants only).

3. *Domain.* $\alpha = 0$ or $\alpha = 1$, which are included.

Because the resident does not depend on b/c and the condition $(b/c)H_{\min} \geq 1$ is monotone in b/c, the secure set grows with b/c. Upper boundaries rise with b/c, whereas fold boundaries stay fixed (Extended Data Fig. 9b–e).

Proposition S14 (Where the secure interval closes)**.** *Suppose that near a fold (r_L, α_L) of the high branch the upper boundary is determined by Eq. (S24). Then an interval of secure tolerance strengths adjoins the fold if $b/c > u'_R(r_L)/\Delta\Phi(r_L)$ and not if $b/c < u'_R(r_L)/\Delta\Phi(r_L)$. Under Shunning this critical ratio is exactly 2 for every error rate, depth and pair of strengths.*

Proof. At the fold, $\mathscr{R}' u'_R = 1$, so the local security condition $(b/c)\Delta\Phi\mathscr{R}' \geq 1$ reads $b/c \geq u'_R/\Delta\Phi$. Under SH, $u_R(r) = e_2 + \kappa r^2$ and $\Delta\Phi(r) = \kappa r$ (Supplementary Table S2), so $u'_R/\Delta\Phi = 2\kappa r/(\kappa r) = 2$. □

For SJ, $u'_R(r)/\Delta\Phi(r) = 2 - e_1(1-2e_2)/\Delta\Phi(r)$, which is 1.9710 at the fold of the selected $(4,3)$ architecture. For SC, $u'_R/\Delta\Phi = 1$, so a fold never closes the interval for $b/c > 1$. These results explain why the secure intervals of SH close at $b/c = 2$ in Extended Data Fig. 9c, whereas that of the selected SJ architecture is still open, though narrow, at $b/c = 2$. Supplementary Table S8 lists the secure intervals of all architectures at $b/c = 5$.

Worked example: Stern Judging with depths (4,3). At $e_1 = e_2 = 0.02$, $P^+_{C|G} = 0.9608$, $P^+_{D|G} = 0.02$, $P^+_{C|B} = 0.0392$ and $P^+_{D|B} = 0.98$, so $\Delta\Phi(r) = 0.9408(2r-1)$ and $u_R(r) = 0.98 - 1.9008\,r + 1.8816\,r^2$. With $\beta = 1$,

$$\mathscr{R}(u;\alpha) = \frac{u^4 \sum_{m=0}^{4} [\alpha(1-u)]^m}{(1-u)^4 + u^3 - (1-u)^4 u^3}.$$

At $\alpha = 1$, $r^*_R = 0.9999996$ and ALLD is excluded ($\Lambda(1) = -4.000$), but $\Lambda(\delta) > 0$ for $0 < \delta < \delta_c = 0.365299$. At $\alpha = 0.8$, $r^*_R = 0.986509$ and $H_{\min} = H(0) = 0.20366 > c/b$, so every O-DISC mutant is excluded, with critical ratio $(b/c)_{\text{crit}} = 1/H_{\min} = 4.910$. The fold lies at $\alpha_L = 0.5669858$ ($r = 0.851537$) and the upper boundary at $\alpha_U = 0.8036040$ ($r = 0.986906$, where $F'(r) < 0$), so the secure interval is $0.5670 < \alpha \leq 0.8036$. The trade-off at $\delta = 0.1$ illustrates the mechanism:

Tolerance strength	r^*_R	$r^*_X(0.1)$	Saved cost δr^*_R	Lost benefit $(b/c)(r^*_R - r^*_X)$	$\Lambda(0.1)$
$\alpha = 1$	0.9999996	0.999789	0.100000	0.001054	+0.098946
$\alpha = 0.8$	0.986509	0.964412	0.098651	0.110484	−0.011833

Table S8. Secure tolerance intervals at $b/c = 5$ ($\beta = 1$, $e_1 = e_2 = 0.02$). Round brackets denote excluded fold endpoints, and square brackets denote included domain or neutral-invasion endpoints. The last two columns give the resident reputation and the critical benefit-to-cost ratio $1/H_{\min}$ at $\alpha = 0.8$ for the selected architectures. One-step SH and SJ have lost the high-reputation resident at $\alpha = 0.8$.

Norm	One-step $(1,1)$	Selected multi-step	$r^*_R(0.8)$	$(b/c)_{\text{crit}}(0.8)$
SC	(0.431929, 0.966915]	(2,0): [0, 0.807064]	0.989782	4.827
SH	(0.864863, 0.989948]	(3,0): (0.591161, 0.806464]	0.986341	4.835
SJ	(0.852446, 0.988514]	(4,3): (0.566986, 0.803604]	0.986509	4.910
SS	[0, 0.965245]	(1,2): [0, 1]	0.977886	1.955

2.4 Alternative reputation filters

Fig. 5a compares the inertial record with two mechanisms that also buffer isolated negative evidence. In both cases only the conversion of assessments into a binary image changes. The norm, errors, the private-assessment closure, the resident selection (highest root reached from $r = 1$) and Eqs. (2) and (18) with $\mathscr{R}$ replaced by the alternative map are kept. The original studies use different strategy sets, observation protocols and evolutionary dynamics, so these are controlled adaptations, not replications. Their parameters were neither optimized nor matched in memory to the inertial record. Supplementary Table S9 summarizes the five curves of Fig. 5a.

Aggregation. Following Michel-Mata et al.[4], an observer samples M assessments of the donor and assigns a good image if at least Q of them are positive. If each sampled assessment is independently positive with probability u,

$$\mathscr{R}_{\text{agg}}(u;M,Q) = \sum_{j=Q}^{M} \binom{M}{j} u^j (1-u)^{M-j}, \qquad \mathscr{R}'_{\text{agg}}(u) = M \binom{M-1}{Q-1} u^{Q-1}(1-u)^{M-Q}. \tag{S25}$$

We use $(M,Q)=(2,1)$, the rule that forgives one of two negative assessments, so that $\mathscr{R}_{\mathrm{agg}}(u)=1-(1-u)^2=2u-u^2$. Then $r^*_{\mathrm{R}}-r^*_{\mathrm{X}}=2(1-u^*_{\mathrm{R}})x+x^2$ with $x=\delta r^*_{\mathrm{R}}\Delta\Phi^*$, and

$$\Lambda_{\mathrm{agg}}(\delta)=r^*_{\mathrm{R}}\,\delta\Big[1-2\,\frac{b}{c}\,(1-u^*_{\mathrm{R}})\Delta\Phi^*-\frac{b}{c}\,r^*_{\mathrm{R}}(\Delta\Phi^*)^2\delta\Big]. \tag{S26}$$

For SJ at the reference parameters, $r^*_{\mathrm{R}}=0.998188$, $u^*_{\mathrm{R}}=0.957431$ and $\Delta\Phi^*=0.937390$, giving $\Lambda_{\mathrm{agg}}=0.599876\,\delta-4.377595\,\delta^2$ and invasion for $0<\delta<0.137033$. In the error-free all-good limit, $\Lambda_{\mathrm{agg}}=\delta[1-(b/c)\delta]$ and the interval ends at $\delta=c/b$.

Quantitative score. Following Schmid et al.[7], an observer keeps an integer score $z\in\{-z_{\max},\dots,z_{\max}\}$ for each target, increases it by one after a positive assessment and decreases it by one after a negative assessment (attempts to leave the range leave the score unchanged), and interprets the target as good if $z\geq z_0$. For i.i.d. input $u\in(0,1)$, the score is a reflecting birth–death chain on $j=z+z_{\max}\in\{0,\dots,2z_{\max}\}$. Detailed balance, $\pi_j u=\pi_{j+1}(1-u)$, gives $\pi_j\propto t^j$ with $t=u/(1-u)$, and

$$\mathscr{R}_{\mathrm{score}}(u;z_{\max},z_0)=\frac{\sum_{j=z_{\max}+z_0}^{2z_{\max}}t^j}{\sum_{j=0}^{2z_{\max}}t^j},\qquad \mathscr{R}_{\mathrm{score}}(0)=0,\quad \mathscr{R}_{\mathrm{score}}(1)=1. \tag{S27}$$

At $u=1/2$ the score is uniform and $\mathscr{R}_{\mathrm{score}}=(z_{\max}-z_0+1)/(2z_{\max}+1)$. For $z_{\max}=1$ and $z_0=0$, $\mathscr{R}_{\mathrm{score}}=u/(1-u+u^2)$. We use $z_{\max}=5$ and $z_0=0$. Under SJ at the reference parameters the selected resident has $1-r^*_{\mathrm{R}}\approx 4.6\times10^{-9}$, ALLD is excluded, and O-DISC invades for $0<\delta<0.383206$. The source model starts scores at zero. Selecting the root reached from $r=1$ is a convention of this adaptation, and SC and SH also have lower roots for $z_{\max}=5$.

Table S9. Opportunistic invasion under SJ ($b/c=5$, $e_1=e_2=0.02$, $\beta=1$). δ_c is the upper end of the interval with $\Lambda>0$ (0 if none).

Mechanism	r^*_{R}	$\Lambda(1)$	δ_c
One-step inertia $(1,1)$, $\alpha=1$	0.995307	−3.975	0.008397
Selected $(4,3)$, $\alpha=1$	0.9999996	−4.000	0.365299
Selected $(4,3)$, $\alpha=0.8$	0.986509	−3.946	0
Aggregation, $(M,Q)=(2,1)$	0.998188	−3.778	0.137033
Quantitative score, $z_{\max}=5$, $z_0=0$	0.99999999	−4.000	0.383206

Supplementary Note 3. Finite-population reputation simulations

The simulations test how reputations form when the mean-field closure is not imposed. Strategy counts are fixed throughout every run, so the simulations concern reputation dynamics only and contain no strategy replacement, mutation or imitation. They retain individual latent records, sampling fluctuations, differences among targets of the same strategy and correlations between the records of different observers. The simulations and the theory share the event rules, records and images, but differ in the closure and in population size. The simulations are used to identify architectures that produce discriminating and consistent private reputations, and the strategy dynamics then use the closed theory with these architectures. No quantitative agreement between simulated g_I and mean-field r_I is claimed or used.

Event algorithm. The population has N individuals with fixed counts n_{ALLC}, n_{ALLD} and n_{DISC}. Each DISC individual i stores a latent record L_{ij} for every individual j, including itself, and all records start in G. Records of ALLC and ALLD individuals are not stored. These strategies never consult reputations, and observers are drawn from the DISC individuals, so omitting them changes no event. Each event consists of the following steps.

1. Draw a donor uniformly from the N individuals and a recipient uniformly from the other $N-1$. Draw an observer uniformly from the n_{DISC} DISC individuals, independently of donor and recipient. The observer may coincide with either.

2. Determine the intended action, C for ALLC and D for ALLD. A DISC donor samples an image from its record of the recipient, Bernoulli$[\sigma(L_{\mathrm{donor,recipient}})]$, and intends C if the image is G.

3. If the intended action is C, replace it by D with probability e_1.

4. The observer samples its own image of the recipient, Bernoulli$[\sigma(L_{\mathrm{observer,recipient}})]$, independently of step 2, and draws a prescribed assessment that is positive with probability ν_{AY} for the realized action A and its image Y. The assessment is reversed with probability e_2.

5. Apply T^+ or T^- (Supplementary Table S3) to $L_{\text{observer,donor}}$. No other record changes.

One sweep consists of $n_{\text{DISC}}N$ events, the number of stored records, so each record is updated once per sweep on average. When the DISC donor is itself the observer, its two consultations in steps 2 and 4 are independent draws from the same record, and the update occurs after both. The instantaneous panels of Fig. 2a–d use the one-step state space with $\alpha = \beta = 0$, which generates the same image process as $(k_1, k_2) = (0, 0)$ (Lemma S1).

Estimators. At a sampled time, let $s_{ij} = \sigma(L_{ij})$, the probability that observer i currently interprets target j as good. Using these probabilities, rather than sampled images, removes sampling noise of the images from the estimates. The mean goodness g_j and the disagreement d_j of Eq. (19) can be computed efficiently as

$$d_j = \frac{(n_{\text{DISC}} - 1)\mu_{1,j} - \mu_{1,j}^2 + \mu_{2,j}}{\binom{n_{\text{DISC}}}{2}}, \qquad \mu_{1,j} = \sum_i s_{ij}, \quad \mu_{2,j} = \sum_i s_{ij}^2, \tag{S28}$$

where i ranges over the DISC observers (including j itself if j uses DISC). This form follows from $\sum_{i<i'}(s_i + s_{i'} - 2s_i s_{i'}) = (n_{\text{DISC}} - 1)\mu_1 - (\mu_1^2 - \mu_2)$. For deterministic images with a fraction x_j of observers interpreting j as good, $d_j = 2n_{\text{DISC}}x_j(1 - x_j)/(n_{\text{DISC}} - 1)$, the finite-observer version of $2x_j(1 - x_j)$. The strategy-level quantities g_I and d_I average g_j and d_j over the n_I targets of strategy I and over sampling times, and they enter Δg and $\bar{d}$ through Eq. (20). Δg measures how differently cooperative strategies and unconditional defectors are treated and is not a classification accuracy. Low $\bar{d}$ alone is not evidence of good reputation formation, because observers can agree that everyone is bad (one-step SH in Fig. 2b). The two statistics are therefore interpreted jointly.

Runs, replication and uncertainty.

- *Image matrices (Fig. 2a–d).* $N = 90$ (30 per strategy), 1,000 sweeps and seed 2027 for each panel. Each matrix shows the final image probabilities s_{ij} of the DISC observers (rows) for all targets (columns, grouped by strategy), which are 0 or 1 for the displayed settings. These are illustrative single realizations.
- *Depth maps (Fig. 2e–h).* $N = 150$ (50 per strategy), $k_1, k_2 \in \{0, \ldots, 6\}$ and $\alpha = \beta = 1$, with one run per condition (base seed 2027). Runs last 1,000 sweeps and are sampled every 5 sweeps from sweep 800 to 1,000 inclusive (41 samples).
- *Replicated analysis (Extended Data Fig. 1).* The same 196 conditions with 30 independent runs each (5,880 runs). Condition seeds are generated by NumPy SeedSequence from the base seed 20260918, the norm index (SC, SH, SJ, SS), k_1, k_2 and the run index. Point estimates are means of the 30 run means, and pointwise 95% percentile intervals use 2,000 bootstrap resamples of run means (seed 2026091801). Samples within a run are never treated as independent.
- *Drift criterion.* For each run, the difference between the means of the second and first halves of its sampling window is computed for Δg and $\bar{d}$, and these differences are bootstrapped across runs. A condition passes if both 95% intervals lie inside $(-0.01, 0.01)$. Conditions that failed were rerun for 3,000, 10,000 and then 50,000 sweeps, with the first 80% of each run discarded and the five-sweep interval retained, until they passed. Only SJ $(4, 5)$ (3,000 sweeps) and SJ $(3, 5)$ (50,000 sweeps) required extension, and all 196 conditions passed. This is a practical check of the sampling window, not a proof of mixing.

Selection of representative architectures. Among conditions that pass the drift criterion and satisfy $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$, we choose the smallest total depth K, then the larger Δg, the smaller $\bar{d}$, the smaller k_1 and the smaller k_2. This selects SC $(2, 0)$, SH $(3, 0)$, SJ $(4, 3)$ and SS $(1, 2)$ (Supplementary Table S10). The cutoffs are operational criteria, not theoretical thresholds, and selection uses no evolutionary or invasion outcome. The stars in Fig. 2e–h apply the same cutoffs and tie-breaking rule to the single-run maps (without the drift criterion) and select the same four pairs. Extended Data Fig. 1e–h repeats the selection for minimum separations $0.70, 0.75, \ldots, 0.90$ and maximum disagreements $0.050, 0.075, \ldots, 0.150$.

Table S10. Selected architectures (30 runs, $N = 150$). Brackets give pointwise 95% bootstrap intervals.

Norm	(k_1, k_2)	Δg	$\bar{d}$
SC	$(2, 0)$	0.9195 [0.9192, 0.9198]	0.0644 [0.0641, 0.0646]
SH	$(3, 0)$	0.9059 [0.9056, 0.9062]	0.0694 [0.0691, 0.0696]
SJ	$(4, 3)$	0.8806 [0.8801, 0.8812]	0.0812 [0.0809, 0.0815]
SS	$(1, 2)$	0.9174 [0.9171, 0.9177]	0.0562 [0.0560, 0.0564]

Supplementary Note 4. Numerical methods

Reputation map and roots. $\mathscr{R}$ is evaluated from Eq. (S6), with arguments clipped to $[0,1]$ and the special case $\alpha=\beta=0$ evaluated as $\mathscr{R}(u)=u$. $\mathscr{R}'$ is evaluated from Eq. (S11) or by central differences. The closed form was checked against the stationary vector of $T(u)$ computed by linear algebra for random depths, strengths and inputs. Homogeneous-DISC roots are found by evaluating F on a uniform grid (2,001 points for the drift curves and the depth planes, 401 points for the 161×161 (α,β) plane and 4,001 points in Fig. 5), refining every sign change by bracketed root finding and classifying roots as stable or unstable when $|F'|>10^{-7}$ and as near-fold otherwise. In Extended Data Fig. 9, candidate roots are the real roots of the polynomial numerator of F (from a Chebyshev fit) and are refined by Newton's method.

Reputations at mixed compositions. Equation (13) is integrated from $r_I=1$ by fourth-order Runge–Kutta steps, with every stage clipped to $[0,1]$. The strategy engine uses step 0.25 until $\max_I|\mathscr{R}(u_I)-r_I|\le10^{-11}$ (fallback step 0.1). The simplex fields of Fig. 3 use steps 0.1 (up to 2×10^4 steps) and then 0.2 (up to 8×10^4 and a further 2×10^5 steps if needed), with residual $\le10^{-12}$.

Residents for local invasion. The scalar flow $\mathrm{d}r/\mathrm{d}\tau=F(r)$ is integrated from $r=1$ by RK4 with step 0.18, to residual 10^{-11} (Extended Data Figs. 5, 8 and 6i–l) or 10^{-9} (Extended Data Fig. 7). The root is then refined by Newton steps of at most 10^{-6} to residual $\le10^{-13}$. The step bound keeps the branch reached by relaxation. This refinement matters where the resident reached from all-good is the all-bad state ($e_2=0$ under SC and SH), at which ALLD is exactly neutral. Local outcomes are classified from λ_{ALLD} and λ_{ALLC} (with $c=1$). ALLD is not excluded if $\lambda_{\mathrm{ALLD}}\ge-10^{-10}$ (this class includes neutral ALLD). Otherwise, ALLC invades if $\lambda_{\mathrm{ALLC}}>10^{-10}$, and the residents resist both mutants if neither condition holds. Dots in Extended Data Fig. 8 mark the last class, and solid segments in Extended Data Fig. 5 mark $R_{\mathrm{L}}<b/c<R_{\mathrm{U}}$.

Strategy trajectories. The engine tabulates the all-good-selected reputations exactly at the 7,381 nodes of a triangular lattice with 120 subdivisions, node (i,j) being $(f_{\mathrm{ALLC}},f_{\mathrm{ALLD}})=(i,j)/120$. A cell is flagged if any reputation component varies by more than 0.02 among its three nodes. In unflagged cells reputations are interpolated barycentrically. In flagged cells, which include every cell crossed by a fold or a steep transition, they are computed exactly by relaxation at the requested composition. Payoffs and the replicator field are computed from these reputations. Earlier pipelines interpolated the replicator field itself on a coarse mesh across folds, which biased sampled cooperation. This was corrected. Trajectories are integrated by the explicit midpoint rule with $\Delta t=0.2$, renormalizing $\boldsymbol{f}$ to the simplex after each stage. After every block of 500 steps (100 time units), ρ_{int} is compared with its value after the previous block. Integration stops after three consecutive changes of at most 10^{-7}, or after 400 blocks (4×10^4 time units).

Validation. (i) Refined settings (160 subdivisions, flag threshold 0.01, $\Delta t=0.1$, 1,000-step blocks, tolerance 10^{-8}) changed $\bar\rho$ by at most 1.3×10^{-5} and changed no outcome class. (ii) An independent implementation that recomputes reputations exactly at every step, without a lattice, reproduced the outcome of every tested trajectory. (iii) The analytic SC no-inertia case is reproduced. (iv) The time cap was reached only on the neutral line of SC without inertia (Proposition S6) and by a few trajectories at b/c close to R_{L} or R_{U}, where an invasion payoff is close to zero. The outcome-gap check (Supplementary Note 1.5) holds for every condition.

Initial compositions. *Shifted Halton* (Fig. 4 and Extended Data Figs. 2e–h, 6e–h and 8). Points $i=1,\dots,512$ of the Halton sequence in bases 2 and 3 are shifted modulo 1 by a uniform random vector drawn with seed 2026. *Scrambled Sobol* (Extended Data Fig. 5). Eight independently scrambled two-dimensional Sobol sequences of 512 points are generated with SciPy qmc.Sobol[8] (seeds 2026–2033). A point (x,y) of the unit square with $x+y>1$ is reflected to $(1-x,1-y)$, which maps the square two-to-one onto the triangle with constant density, and $(f_{\mathrm{ALLC}},f_{\mathrm{ALLD}},f_{\mathrm{DISC}})=(x,y,1-x-y)$. The same compositions are used for all conditions within a figure.

Opportunistic invasion. Fig. 5 evaluates $\Lambda(\delta)$ at 5,015 distinct values of δ, namely 4,001 uniform points on $[0,1]$, 1,001 on $[0,0.025]$ and 100 logarithmically spaced points on $[10^{-9},10^{-3}]$. Nonzero sign changes are refined by Brent's method (tolerance 2×10^{-13}). The scan over tolerance strengths uses 1,001 uniform values of α plus points next to the two boundaries (1,006 values). α_{L} is obtained by solving $F=\partial_r F=0$ and α_{U} by Brent's method on Eq. (S24). The full δ-grid confirmed that no finite δ invades inside the interval. Residents of the alternative filters are obtained from $r=1$ by $r\leftarrow0.8r+0.2\mathscr{R}_{\mathrm{X}}[u_{\mathrm{R}}(r)]$ until successive values differ by less than 2×10^{-14}. Extended Data Fig. 9 uses at least 601 values of α, refined near folds, 161 values of $b/c\in[2,8]$ and $\beta=1$. Because $\mathscr{R}$ is rational in u and u_{X} is linear in δ, $H(\delta)$ is a rational function of δ. Its minimum is found by comparing $H(0)$, $H(1)$ and H at every real root in $(10^{-8},1)$ of the numerator of $\mathrm{d}H/\mathrm{d}\delta$ (roots accepted if their imaginary part is below 10^{-7}). A tolerance strength is secure if $1-(b/c)H_{\min}\le10^{-10}$. This is a finite-precision search, not a proof that no extremely narrow interior extremum exists. In all reported cases the minimum was at $\delta\to0$.

Software. Analyses use Python 3.11 or later with NumPy, SciPy, pandas and Matplotlib, a C99 implementation of the strategy engine, and Numba for the finite-population simulations and simplex fields. The verification was run with Python 3.13.17,

NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, Matplotlib 3.11.2 and GCC 13.3. Every notebook loads the supplied data by default and regenerates it when its recompute option is set.

Supplementary Note 5. Notation

Supplementary Table S11 lists every symbol used in the Methods and in this Supplementary Information. Each symbol has a single meaning.

Table S11. Symbols.

Symbol	Meaning
Game, strategies and norms	
b, c	Benefit to the recipient and cost to the donor of realized cooperation; $b > c > 0$.
$I, J \in \{\mathrm{ALLC}, \mathrm{ALLD}, \mathrm{DISC}\}$	Strategies; f_I frequency, $\boldsymbol{f}$ composition, $\sum_I f_I = 1$.
$A \in \{\mathrm{C}, \mathrm{D}\}$	Intended action (cooperate, defect).
$Y \in \{\mathrm{G}, \mathrm{B}\}$	Binary image of a recipient held by an observer (good, bad).
ν_{AY}	Probability that the norm prescribes a positive assessment of realized action A towards a recipient with image Y.
p, q	Norm parameters; SC $(1,0)$, SH $(0,0)$, SJ $(0,1)$, SS $(1,1)$.
e_1, e_2	Implementation error (intended C becomes D) and assessment error (recorded assessment reversed).
ε, κ	$\varepsilon = (1-e_1)(1-e_2) + e_1 e_2$; $\kappa = (1-e_1)(1-2e_2) = \varepsilon - e_2$.
$P^{+}_{\mathrm{A}\mid\mathrm{Y}}$	Probability of a positive recorded assessment given intended action A and the observer's image Y of the recipient (errors included).
$\Phi_{\mathrm{C}}(r)$, $\Phi_{\mathrm{D}}(r)$	Probability that intended C or D is assessed positively when the observer regards the recipient as good with probability r.
$\Delta\Phi(r)$	Assessment contrast $\Phi_{\mathrm{C}}(r) - \Phi_{\mathrm{D}}(r) = \kappa[r + (p-q)(1-r)]$.
Inertial record	
$L_{ij} \in \mathscr{L}$	Latent record of observer i about target j; $\mathscr{L} = \{\mathrm{G}, \mathrm{W}_1, \ldots, \mathrm{W}_{k_1}, \mathrm{B}, \mathrm{S}_1, \ldots, \mathrm{S}_{k_2}\}$.
W_m, S_n	Warning states ($m = 1, \ldots, k_1$) and recovery states ($n = 1, \ldots, k_2$).
k_1, k_2, K	Tolerance depth, strictness depth, total depth $K = k_1 + k_2$.
α, β	Tolerance strength and strictness strength, $0 \le \alpha, \beta \le 1$.
$\sigma(\ell)$, $\boldsymbol{\sigma}$	Probability that state ℓ is interpreted as good; column vector of these probabilities.
u, v	Probability that a recorded assessment of a target is positive; $v = 1 - u$.
T^{+}, T^{-}, $T(u)$	Transition matrices after a positive or negative assessment; $T(u) = uT^{+} + vT^{-}$.
$\pi_\ell(u)$, $Z(u)$	Stationary distribution of a record with i.i.d. input u; normalization $Z = v^{k_1} + u^{k_2} - v^{k_1} u^{k_2}$.
$\mathscr{R}(u)$	Reputation map $\mathscr{R}(u; \alpha, \beta, k_1, k_2) = \sum_\ell \pi_\ell(u)\sigma(\ell)$; $\mathscr{R}' = \mathrm{d}\mathscr{R}/\mathrm{d}u$.
Σ_{W}, Σ_{S}, Ω	Warning-path weight, recovery-path weight and numerator of $\mathscr{R}$.
$\Theta_{\mathrm{G}\to\mathrm{B}}$, $\Theta_{\mathrm{B}\to\mathrm{G}}$	Number of assessments needed to leave good (bad) standing, starting from G (B).
Population reputations	
r_I, $\bar{r}$	Probability that an observer regards a type-I individual as good; $\bar{r} = \sum_I f_I r_I$.
h_{GG}, h_{GB}, h_{BB}	Probabilities that donor and observer both regard a random recipient as good, hold a given ordered disagreement, or both regard it as bad.
u_I	Probability that a type-I donor is assessed positively.
$\boldsymbol{\pi}_I$	Row vector of latent-state probabilities of records about type-I targets.
τ, t', t	Auxiliary relaxation time; time in assessments per record; evolutionary time.
$u_{\mathrm{R}}(r)$, $F(r)$	Resident assessment probability $r\Phi_{\mathrm{C}}(r) + (1-r)\Phi_{\mathrm{D}}(r)$ in a homogeneous DISC population; $F(r) = \mathscr{R}[u_{\mathrm{R}}(r)] - r$.
r^{*}_{R}, u^{*}_{R}, Φ^{*}_{A}, $\Delta\Phi^{*}$	Selected resident reputation (highest strictly stable root); $u_{\mathrm{R}}(r^{*}_{\mathrm{R}})$; $\Phi_A(r^{*}_{\mathrm{R}})$; $\Delta\Phi(r^{*}_{\mathrm{R}})$.
Evolution and cooperation	
Π_I, $\bar{\Pi}$	Expected payoff of strategy I; population mean payoff.
ρ_{int}, ρ	Intended and realized cooperation rate, $\rho = (1-e_1)\rho_{\mathrm{int}}$.
$\bar{\rho}$, ω_{C}, $\bar{\rho}_{\mathrm{C}}$	Mean long-run cooperation over initial compositions; cooperative-basin size; cooperation within the basin.
n_{init}	Number of initial compositions (512 per sample).
$f^{\dagger}$	DISC frequency of the line of equilibria under SC without inertia.
Invasion	
χ_{X}	Probability that a rare mutant X intends to help a resident recipient.
u_{X}, r^{*}_{X}	Positive-assessment probability and stationary reputation of the mutant.

Table S11. Symbols (continued).

Symbol	Meaning
Λ_X	Normalized payoff advantage $(\Pi_X - \Pi_R)/[c(1-e_1)]$.
λ_{ALLD}, λ_{ALLC}	Unnormalized payoff advantages of rare ALLD and ALLC over residents.
r^*_{ALLD}, r^*_{ALLC}	Reputations of rare ALLD and ALLC in the resident population, $\mathscr{R}(\Phi^*_D)$ and $\mathscr{R}(\Phi^*_C)$.
R_L, R_U	Lower and upper benefit-to-cost thresholds for resistance to ALLD and ALLC.
H_{ALLD}, H_{ALLC}	Reputation lost per unit of withheld cooperation (ALLD) and gained per unit of extra cooperation (ALLC).
δ	Probability that O-DISC withholds an intended cooperation.
$\Lambda(\delta)$, $H(\delta)$	Normalized O-DISC advantage; reputation lost per unit of withheld cooperation.
H_{min}, δ_c	$\min_{0\le\delta\le1} H(\delta)$; upper end of the interval with $\Lambda > 0$.
α_L, α_U	Lower (fold) and upper (invasion) boundaries of a secure interval of tolerance strengths.
$\mathscr{R}_{agg}$, M, Q	Aggregation map; number of sampled assessments; required number of positive ones.
$\mathscr{R}_{score}$, z, z_{max}, z_0	Quantitative-score map; score; score bound; good-image threshold.
Finite-population simulations	
N, n_I	Population size and number of type-I individuals.
s_{ij}	$\sigma(L_{ij})$, the probability that DISC observer i currently interprets target j as good.
g_j, g_I	Mean goodness of target j; average over type-I targets and sampling times.
d_j, d_I	Probability that two distinct DISC observers interpret target j differently; average over type-I targets and times.
Δg, $\bar{d}$	Reputational separation and mean disagreement.

Supplementary References

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